Principles of Spatial Analysis

Week 08: Accessibility Analysis



This week

- Transport network and accessibility analysis.
- Spatial network structure.
- Dijkstra’s shortest path algorithm.
- An example of transport network and accessibility analysis.

Spatial interaction

- We have mostly dealt with static events: events that happen at a particular point in time and space.
- Spatial interaction is concerned with the idea that there are relations between different locations (e.g. measured in people or goods travelling).

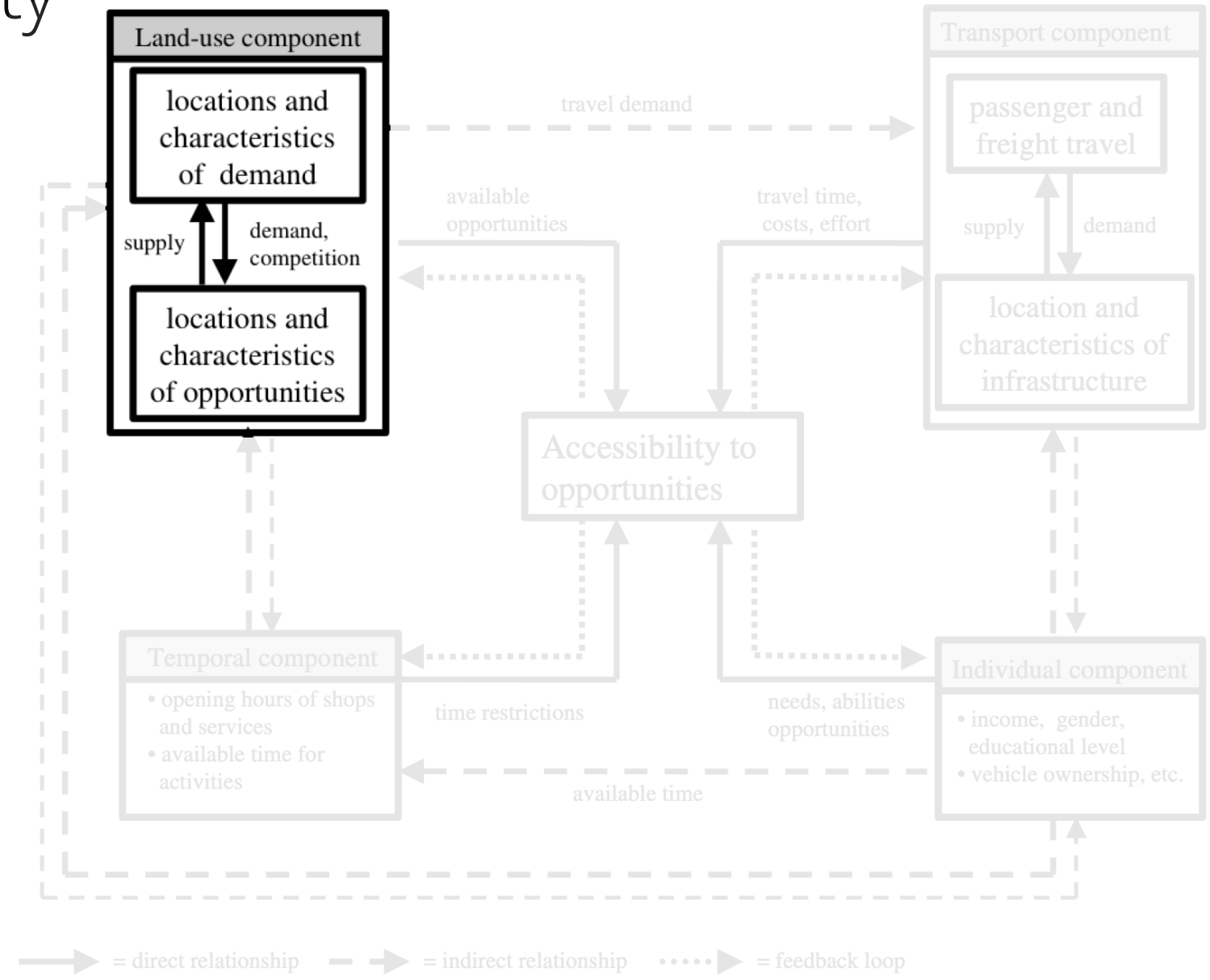
Transport links actors and activities



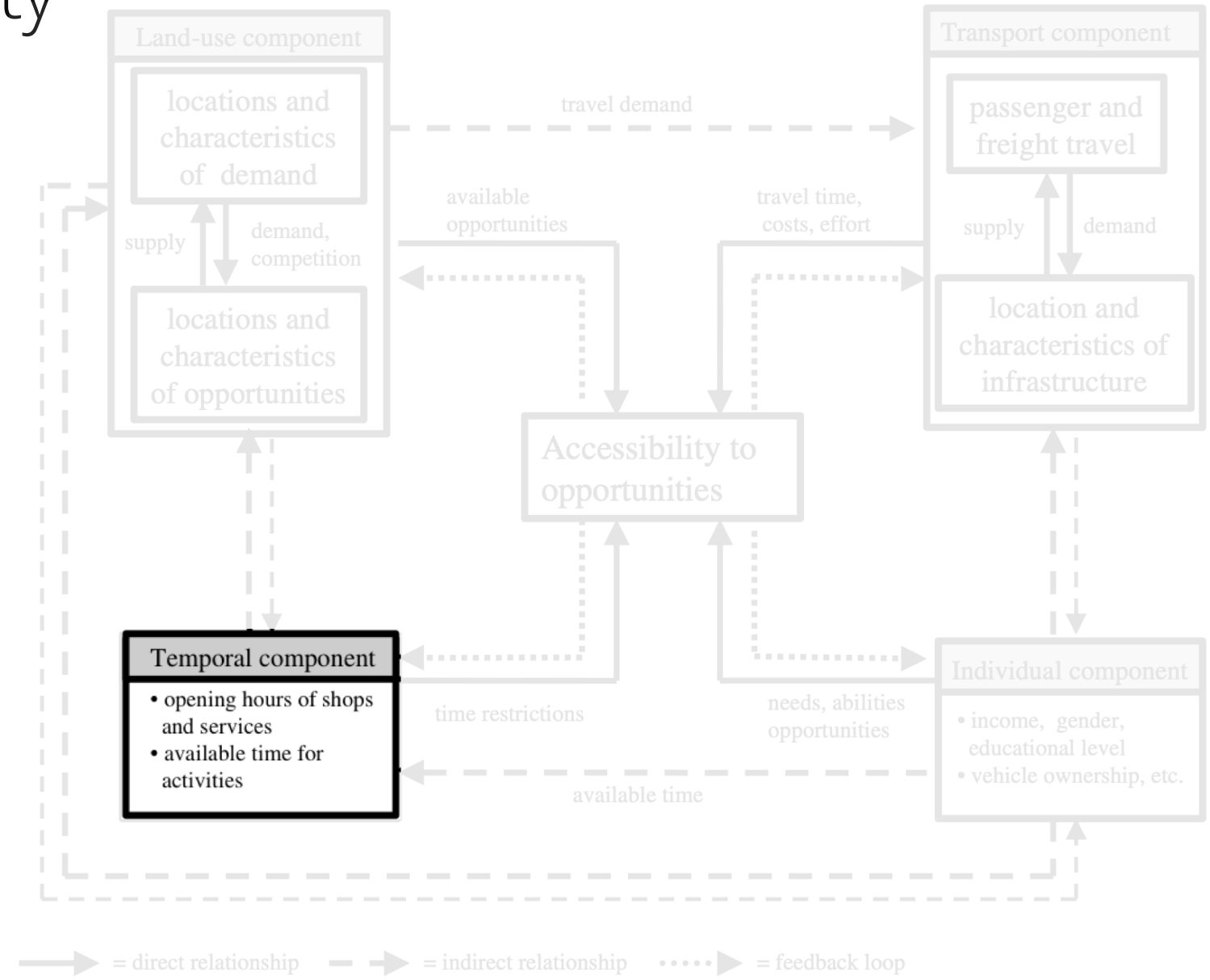
Time geography

- Proposed by Torsten Hägerstrand in the 1970s.
- Time geography describes the life of an individual as a continuous path through time and space, constituted by movements through space and activities localised in space.
- All activities are governed by three constraints: capability constraints, coupling constraints, and authority constraints.
- These constraints can mitigate but also reinforce one another's impacts on activity participation and travel behaviour.
- When people meet their individual space-time paths form a bundle.

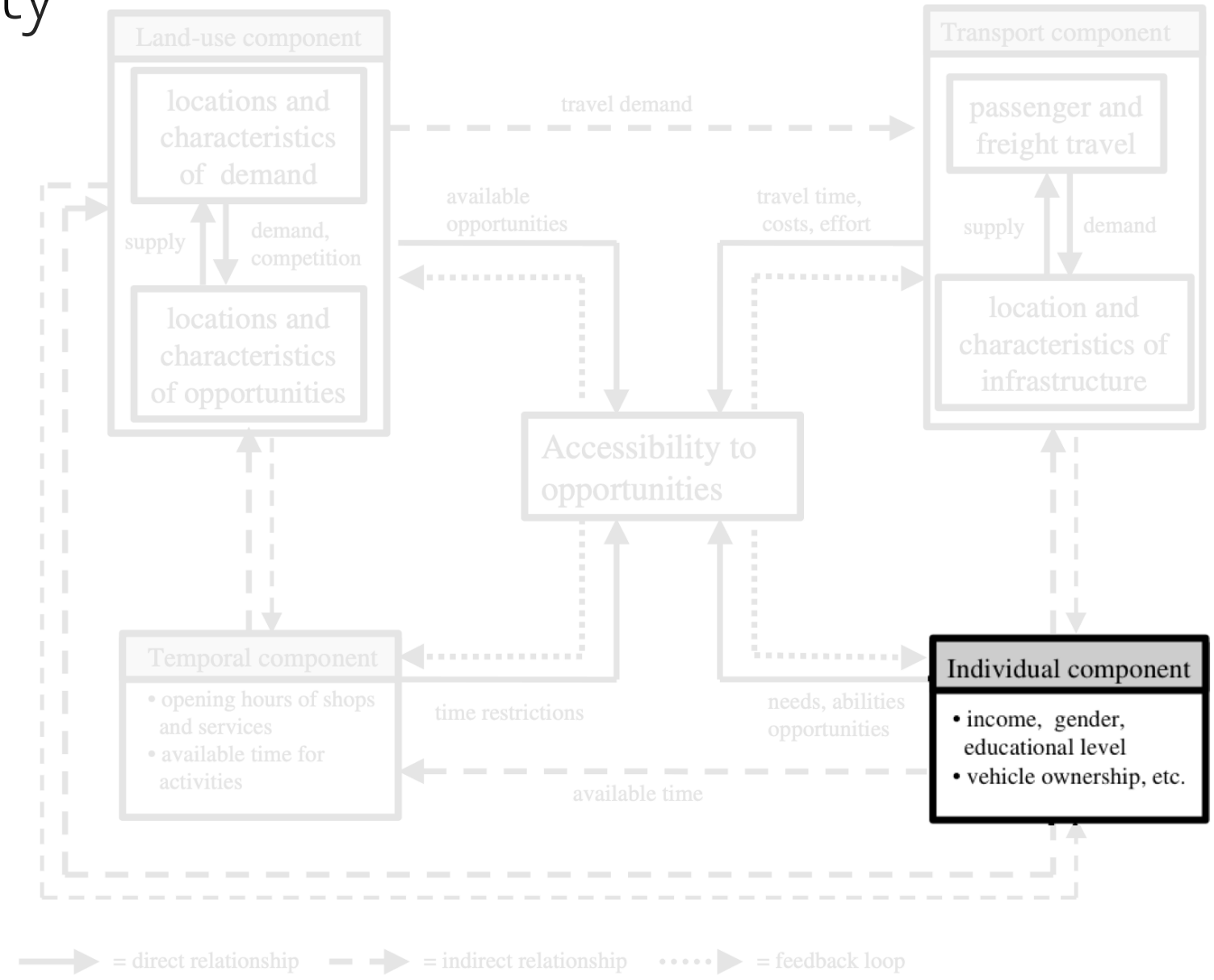
Accessibility



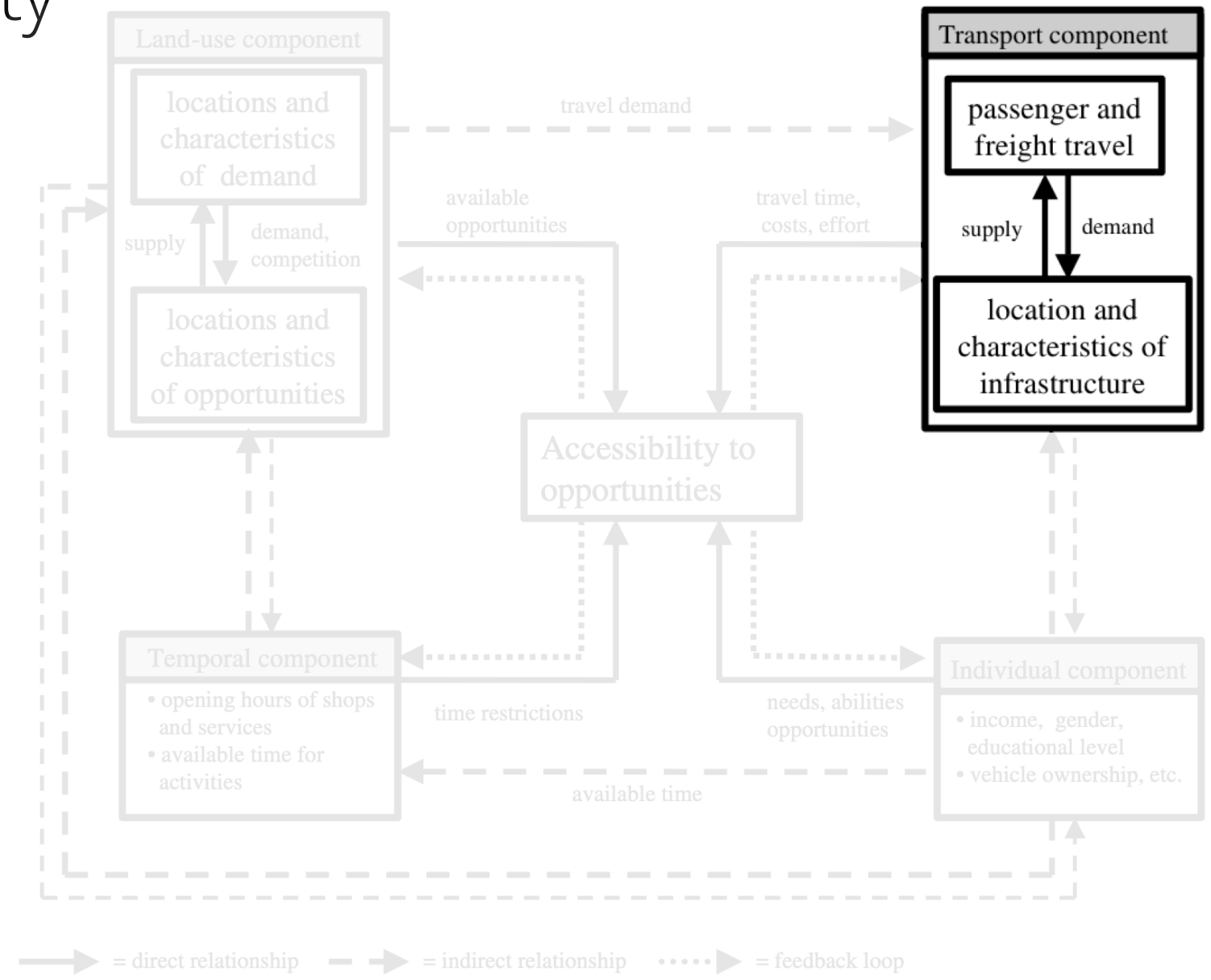
Accessibility



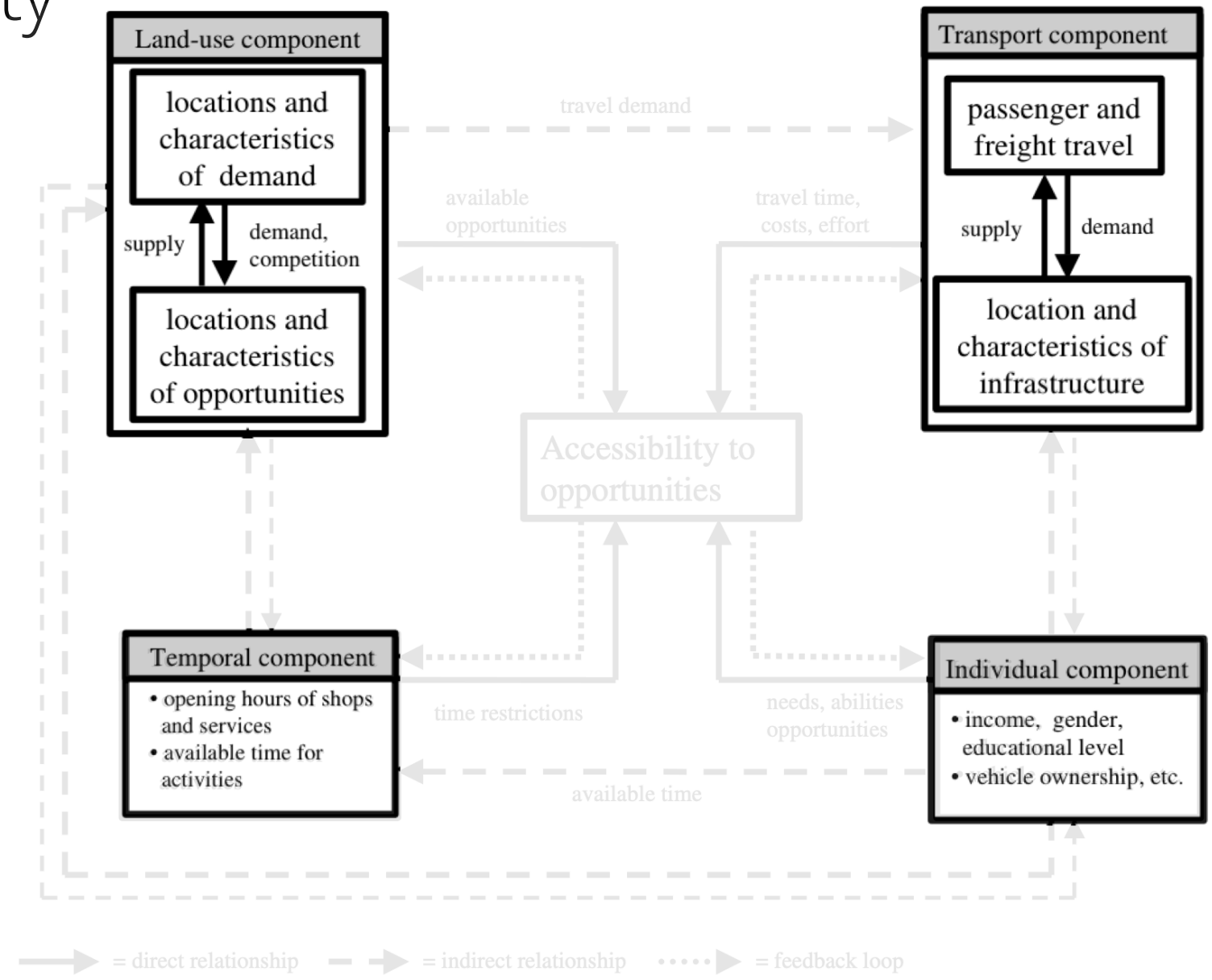
Accessibility



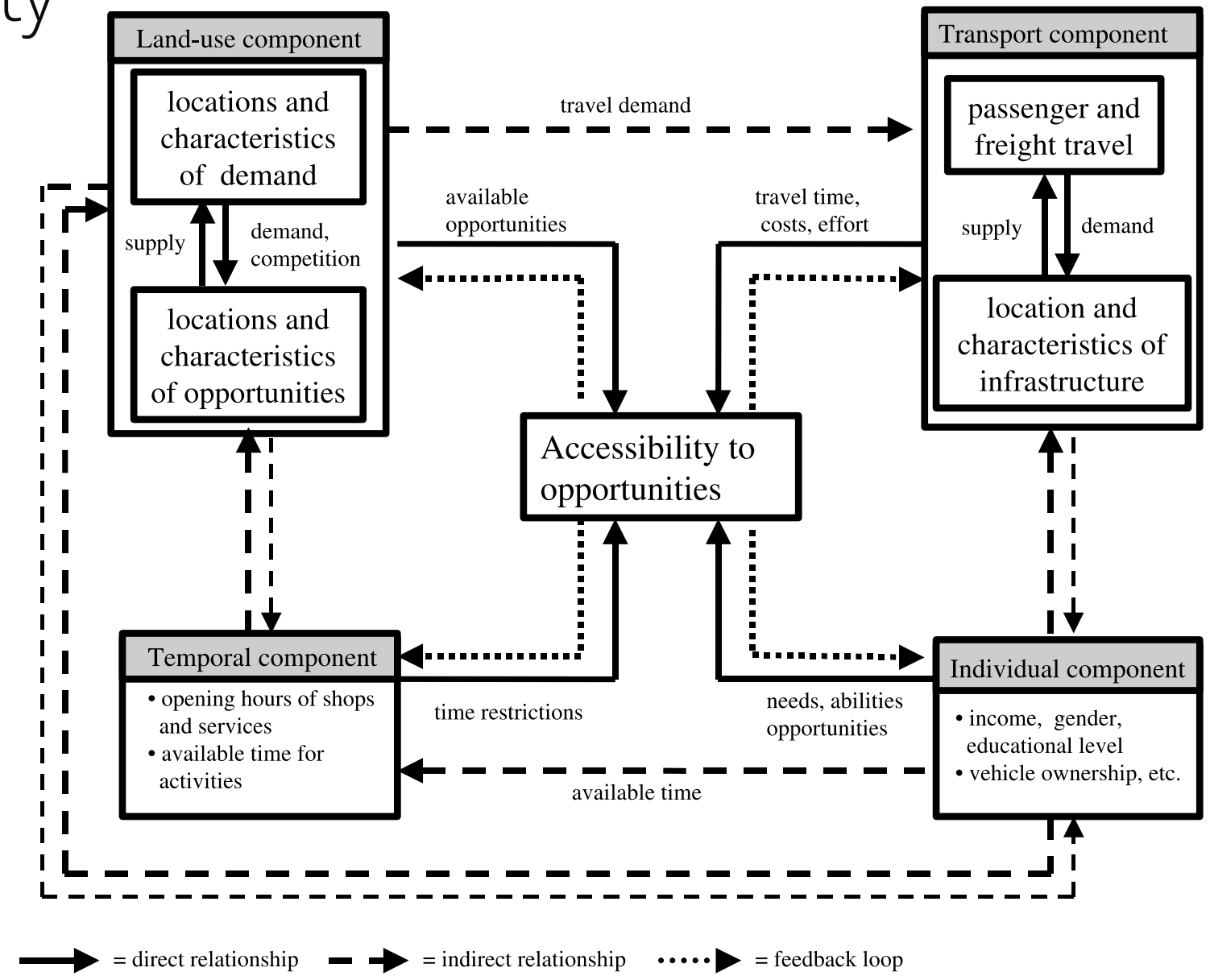
Accessibility



Accessibility

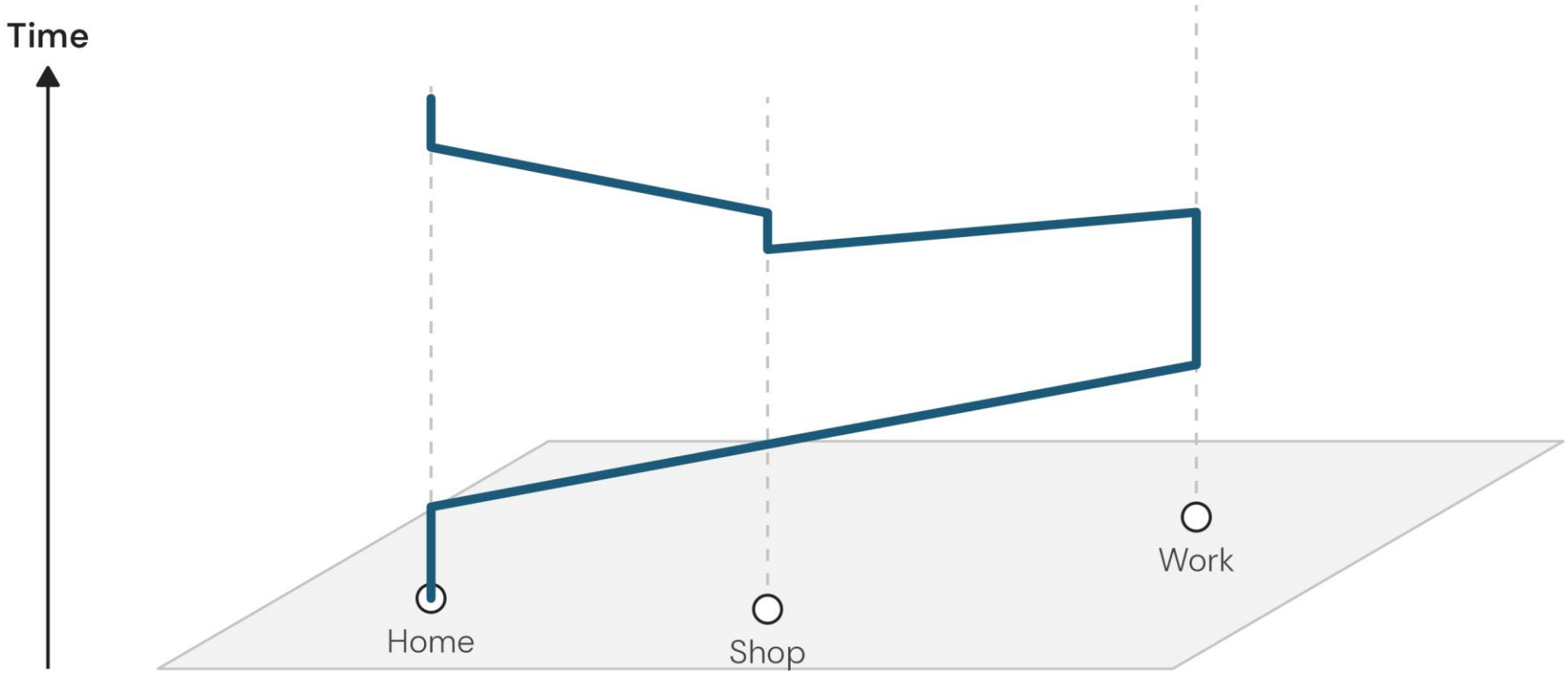


Accessibility

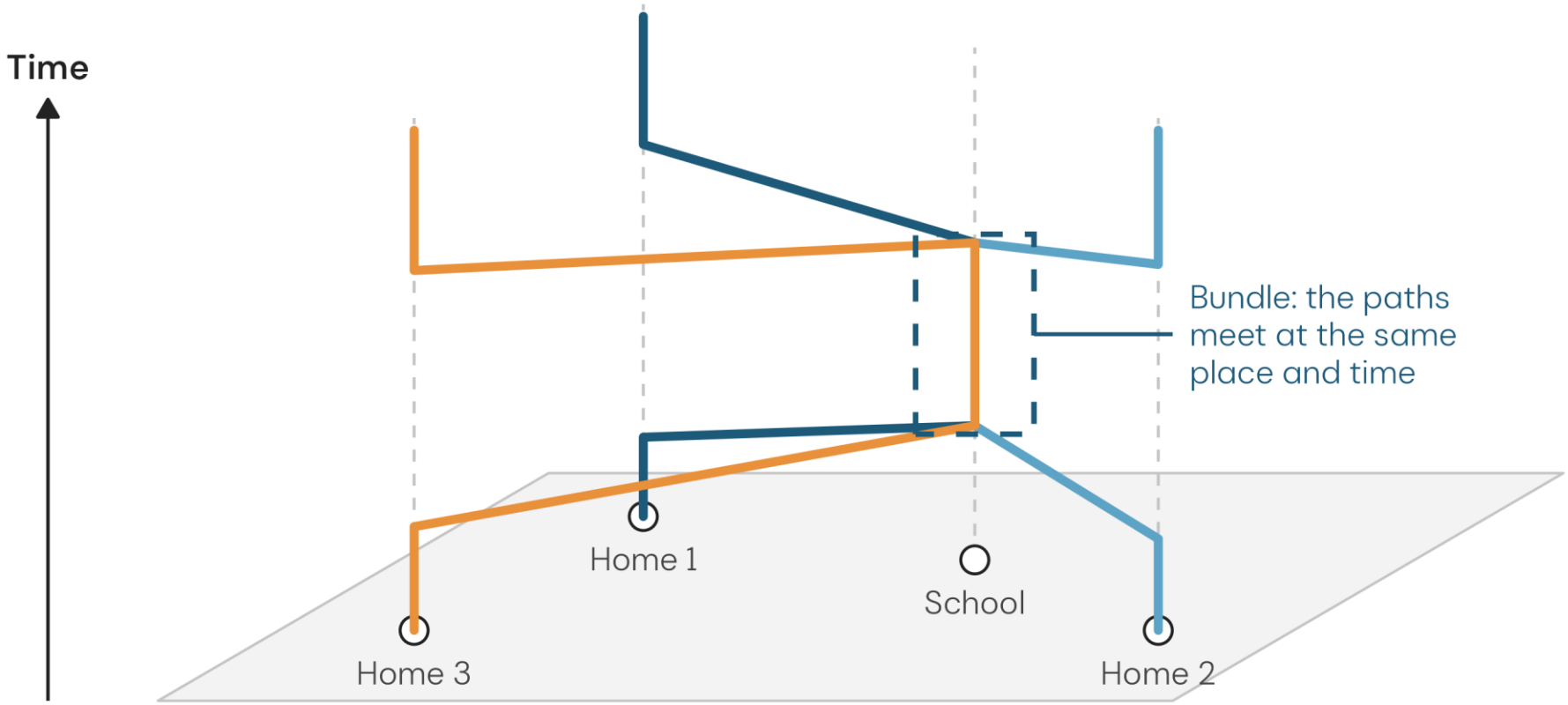


Geurs and Van Wee 2004

Time geography: a path through space and time



Bundles: when paths meet



Typical accessibility questions

Typical questions where accessibility analysis comes in:

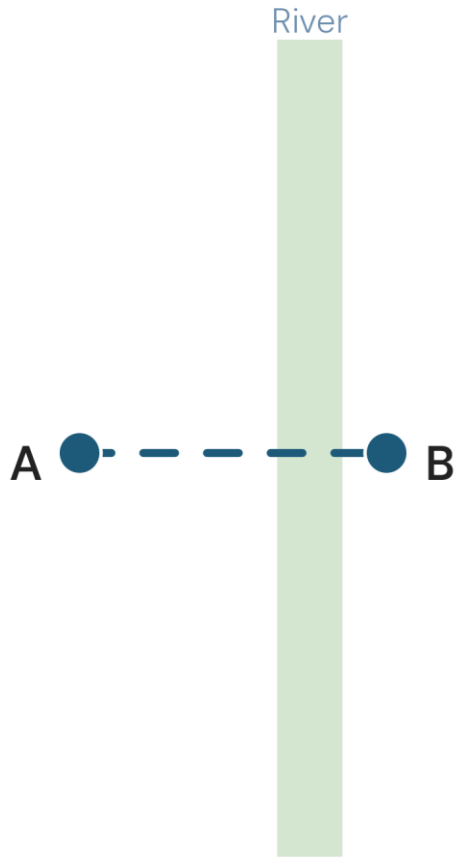
- How many jobs / shops / people can I reach within 15 / 30 / 60 minutes of travel?
- How long do I need to travel to reach N jobs / shops / people?
- How does accessibility differ spatially?
- How does accessibility differ temporally?

What accessibility analysis needs

To quantitatively measure accessibility in a GIS we need at least:

- A set of origins (e.g. homes or schools).
- A set of destinations (e.g. shops, jobs or fast-food outlets).
- Some form of a digital spatial network to connect origins and destinations.

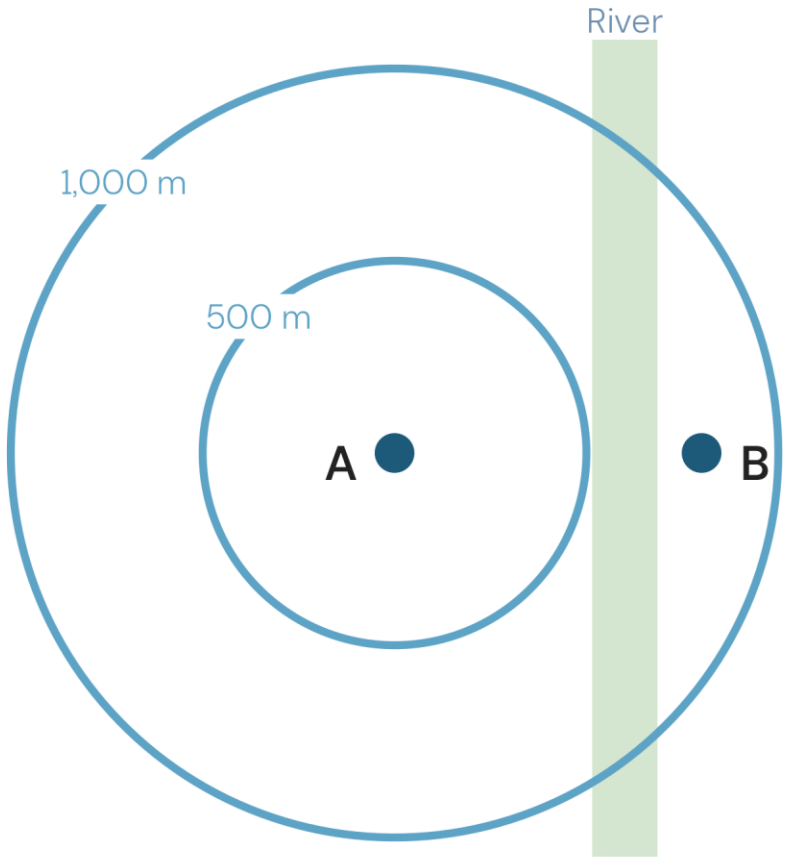
Accessibility of B from A



Straight line from A to B

800 m

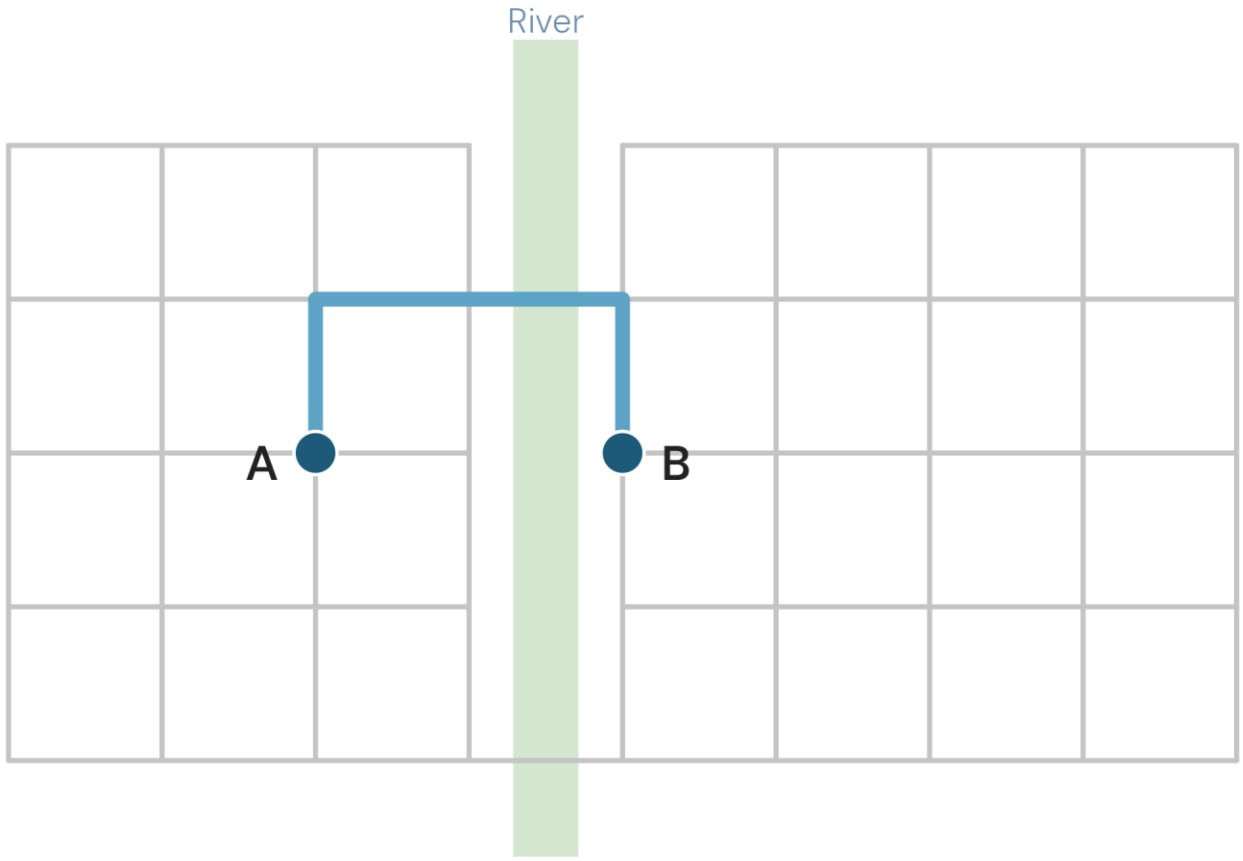
A buffer measures straight-line distance



B lies inside the

1,000 m buffer

The network gives the real distance



Straight line

800 m

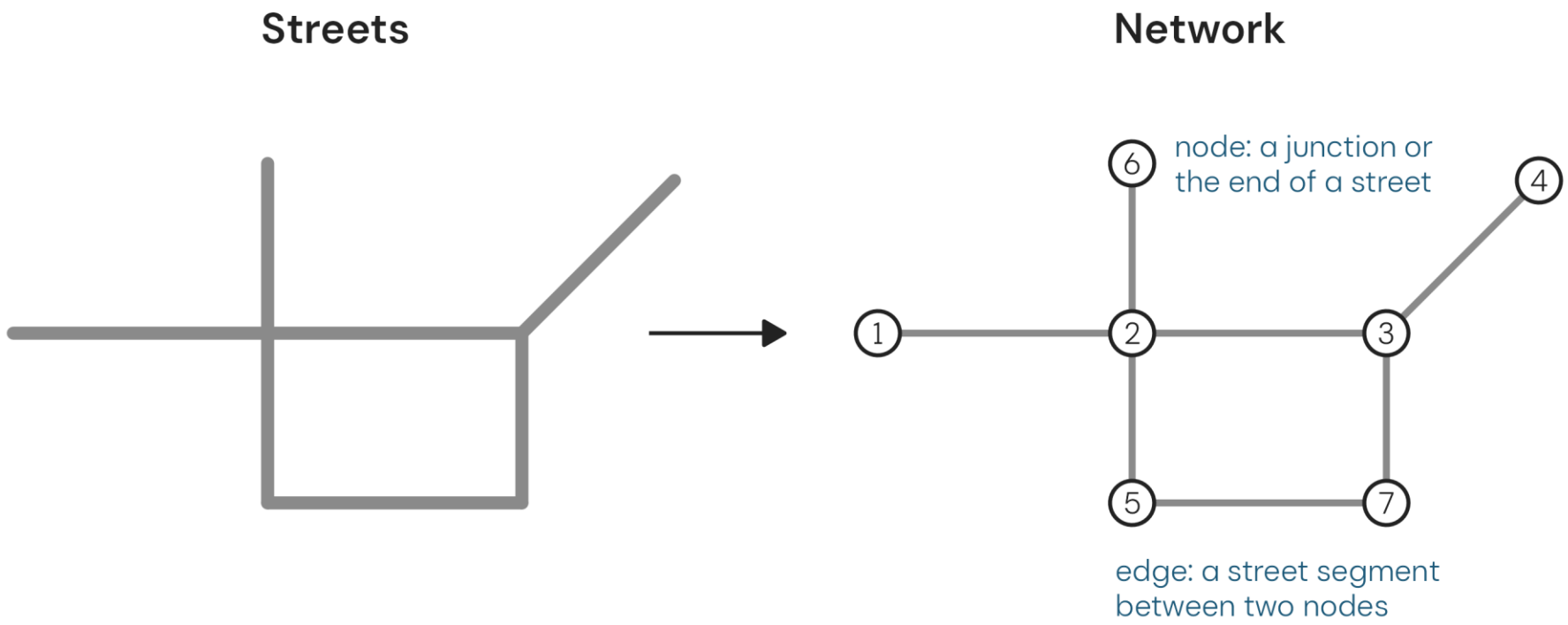
Route along the streets

1,600 m

Spatial network

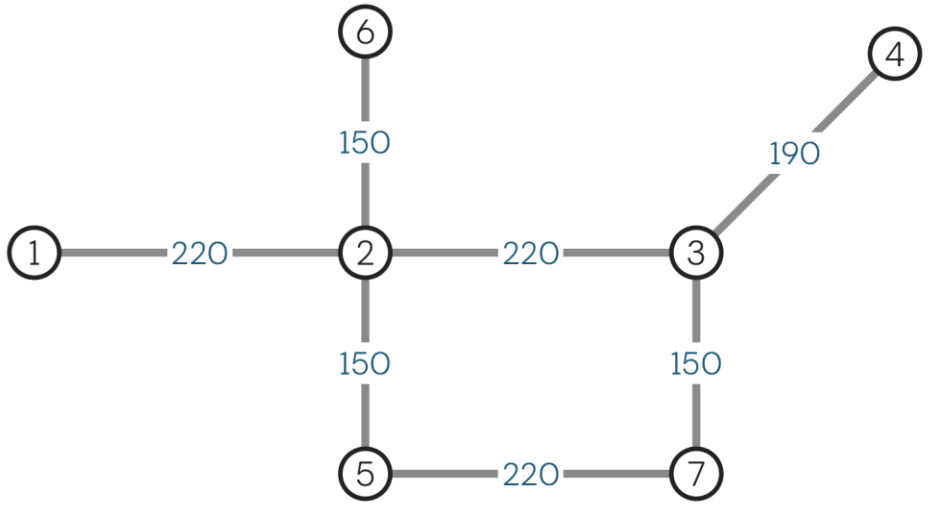
- A spatial network is an organised system or collection of nodes and edges embedded in geographic space.
- Nodes can be a representation of physical objects in geographical space, and edges show what connections are formed between the objects.
- Examples of networks: street configuration, transportation and shipping routes, river basins, telecommunication lines, etc.

A network is nodes and edges



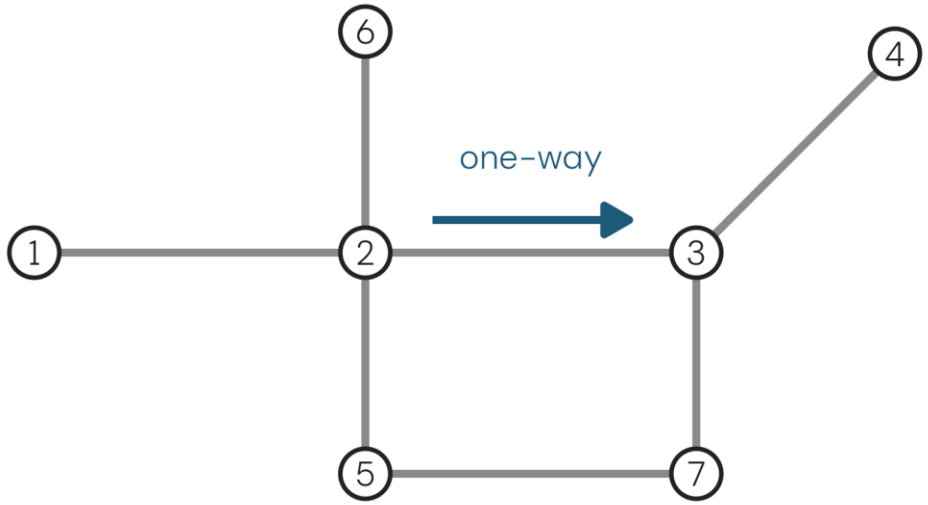
Edges carry a cost

Each edge carries a cost: here its length in metres



Edge	Length (m)	Walk (min)
1 – 2	220	2.9
2 – 3	220	2.9
3 – 4	190	2.5
2 – 6	150	2.0
5 – 7	220	2.9
7 – 3	150	2.0
2 – 5	150	2.0

Edges can have a direction



From 2 to 3

220 m

From 3 back to 2

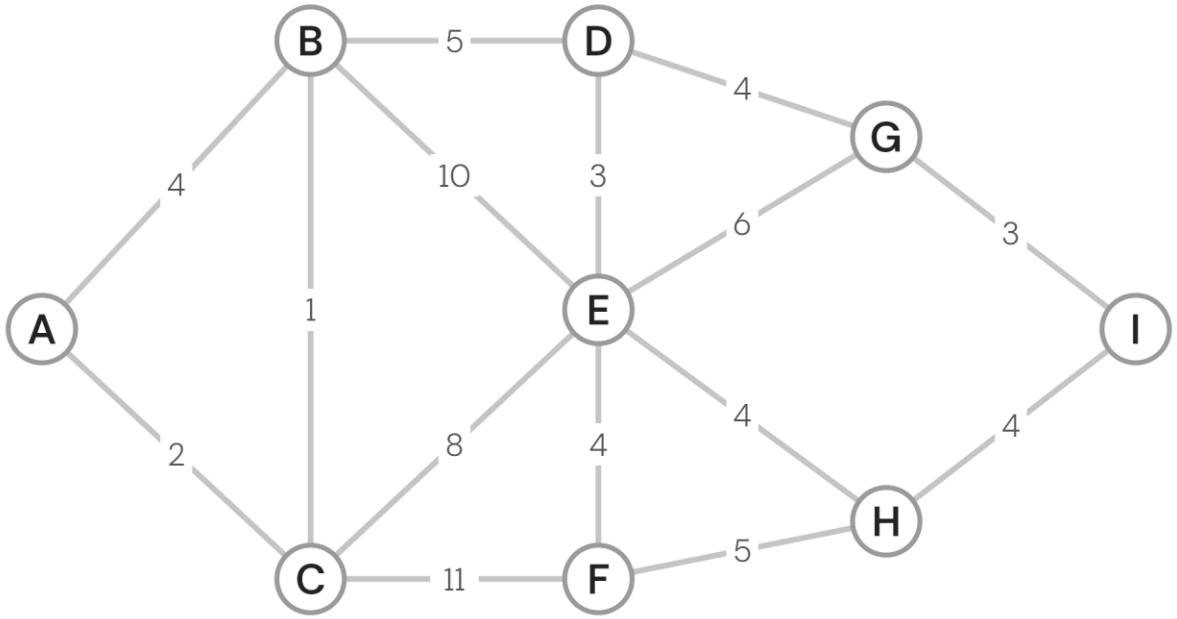
520 m

The way back goes round the block via 7 and 5

Dijkstra’s algorithm: cheapest routes

- Every edge has a cost: distance, travel time or something else.
- Start at the origin with cost 0. Every other node is not yet reached.
- Settle the cheapest node that is not yet settled, then update the costs of its neighbours.
- Stop when the destination is settled, or continue to reach every node.

A network with a cost on every edge

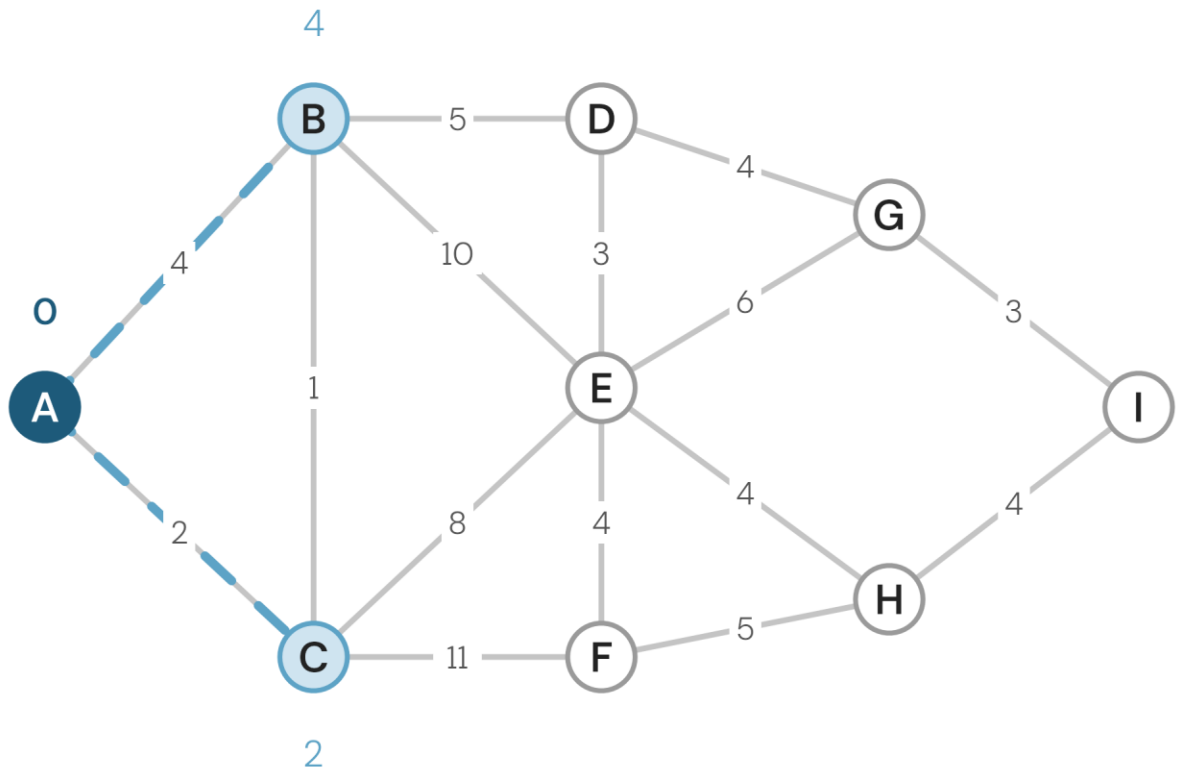


Settled (final cost)

Candidates (best cost so far)

A 0 (the origin)

Step 1: settle the origin, label its neighbours



Settled (final cost)

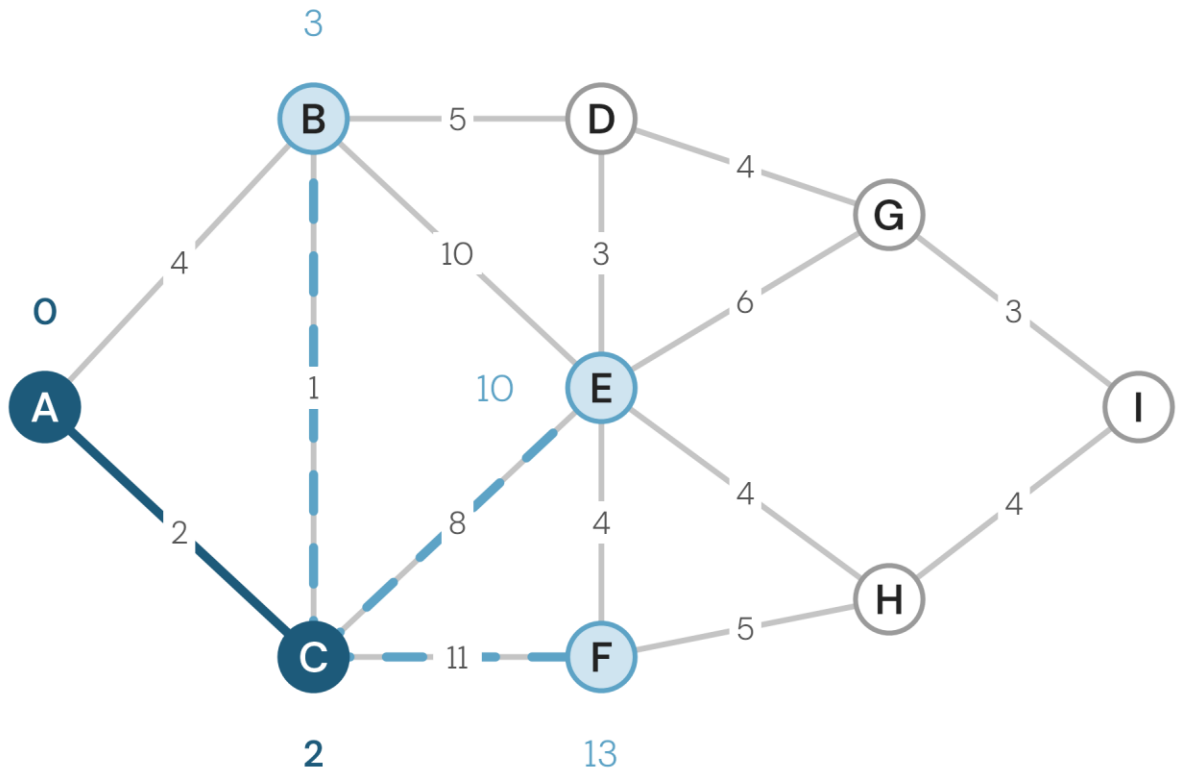
A 0

Candidates (best cost so far)

B 4 (via A)

C 2 (via A)

Step 2: settle the cheapest candidate



Settled (final cost)

A 0

C 2

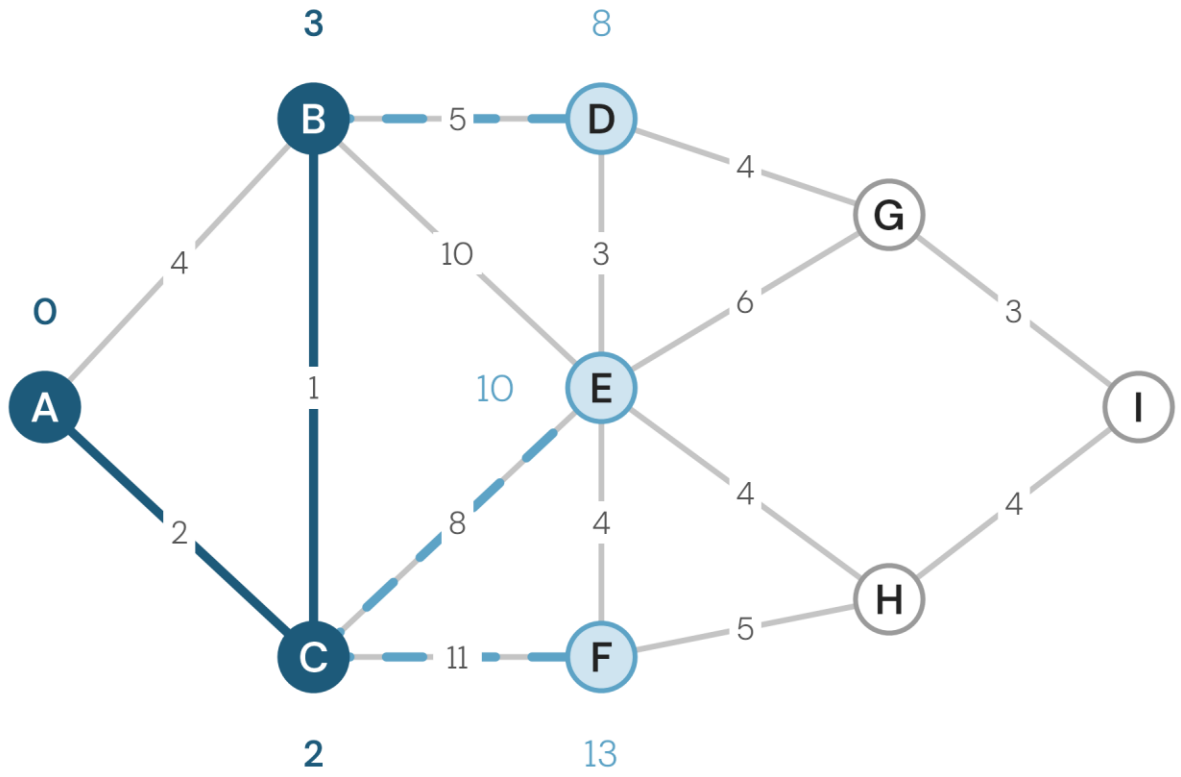
Candidates (best cost so far)

B 3 (via C)

E 10 (via C)

F 13 (via C)

Step 3: repeat with the next cheapest



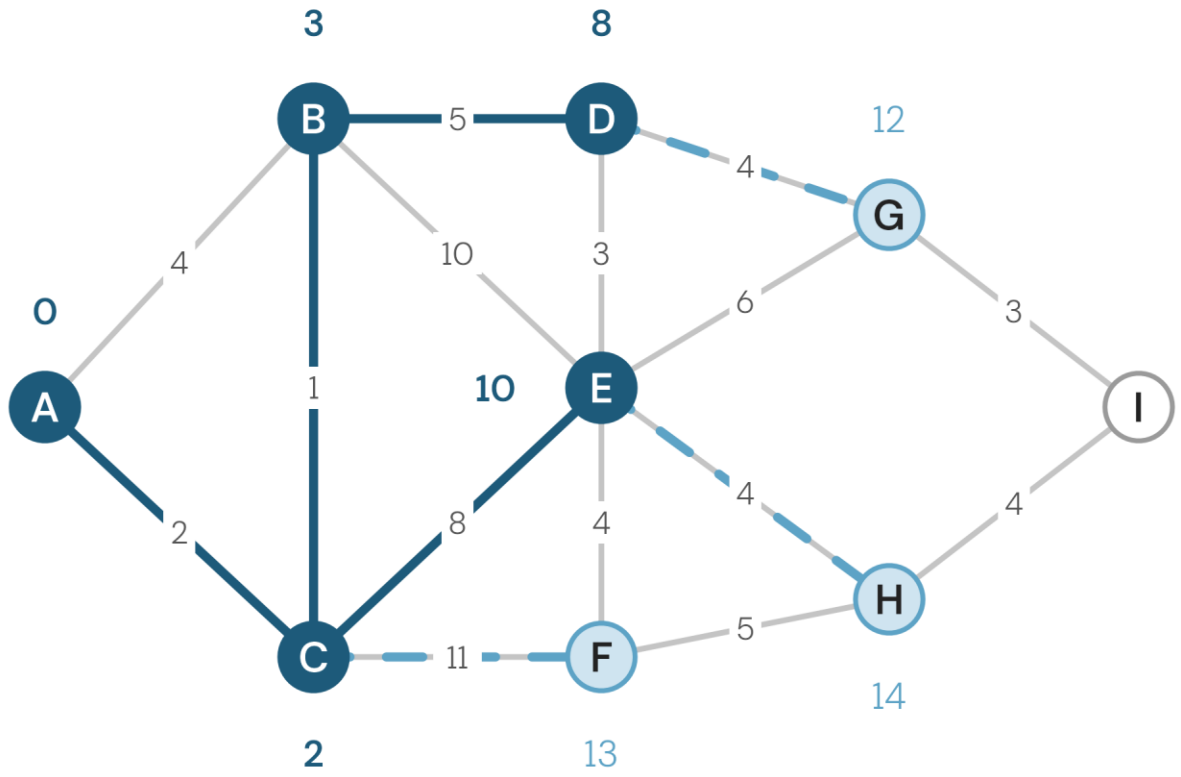
Settled (final cost)

A 0
C 2
B 3

Candidates (best cost so far)

D 8 (via B)
E 10 (via C)
F 13 (via C)

Step 4: keep going



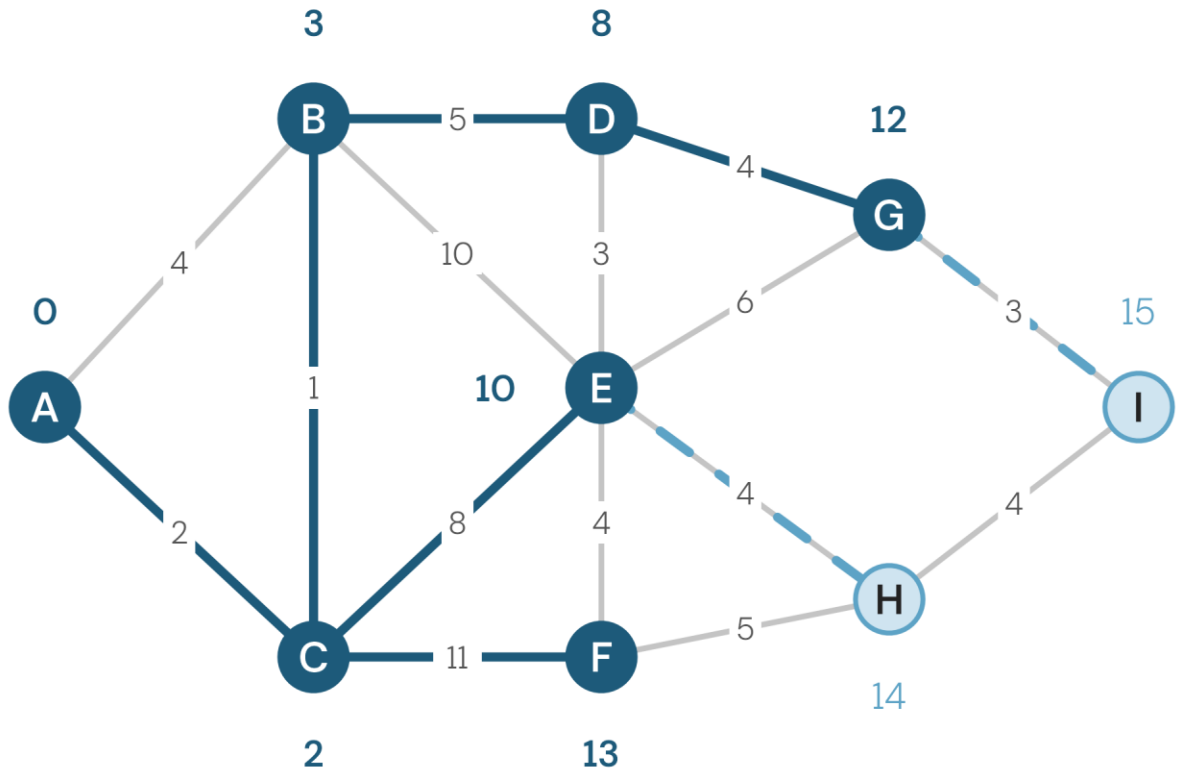
Settled (final cost)

A 0
C 2
B 3
D 8
E 10

Candidates (best cost so far)

F 13 (via C)
G 12 (via D)
H 14 (via E)

A labelled destination is not yet settled



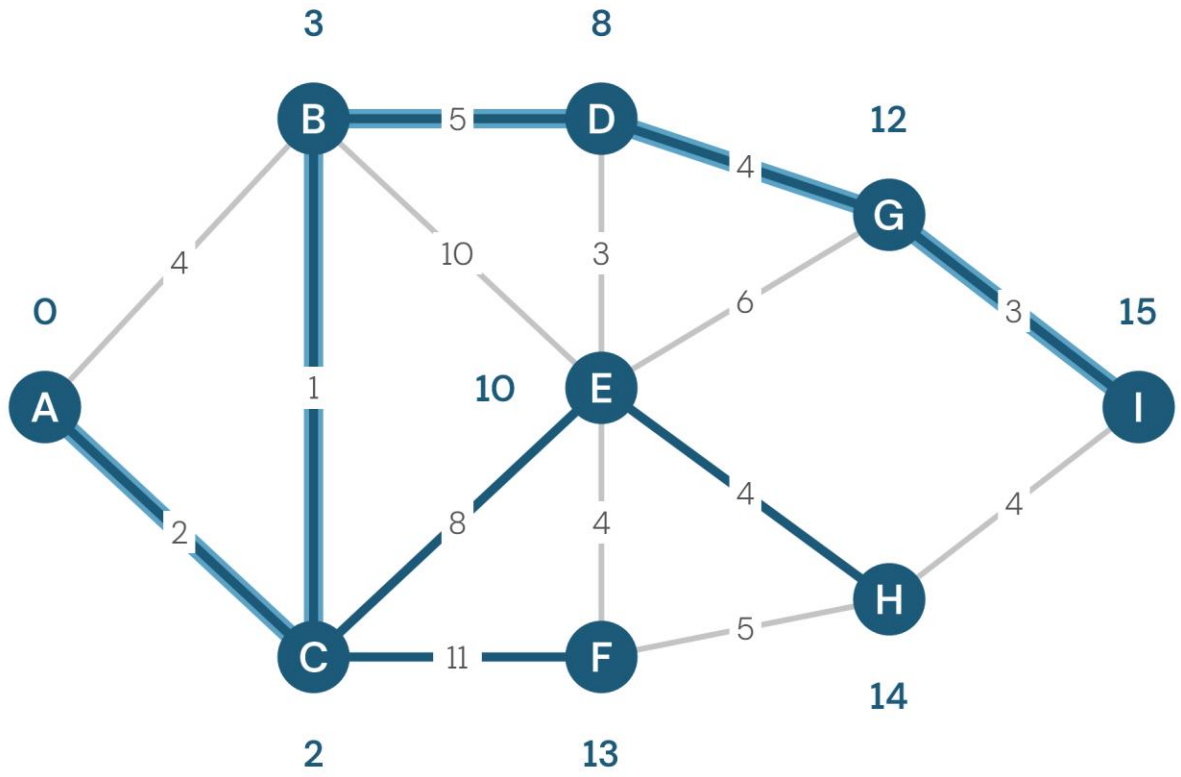
Settled (final cost)

A 0
C 2
B 3
D 8
E 10
G 12
F 13

Candidates (best cost so far)

H 14 (via E)
I 15 (via G)

Result: the cheapest route from A to I costs 15



Settled (final cost)

A 0
C 2
B 3
D 8
E 10
G 12
F 13
H 14
I 15

Candidates (best cost so far)

none left

From distances to an accessibility score

Origins (schools)	Destinations (fast-food outlets) F1	F2	F3	F4	F5	F6	Within 500 m
S1	1,200	**260**	1,480	**200**	**430**	**140**	**4**
S2	**310**	1,390	**480**	970	510	**380**	**3**
S3	1,080	**260**	1,040	**360**	810	1,190	**2**
S4	**370**	**360**	1,410	**450**	990	1,400	**3**

Network distance in metres. Shaded: within 500 m, counted in the score.

Accessibility with Dijkstra: what we need

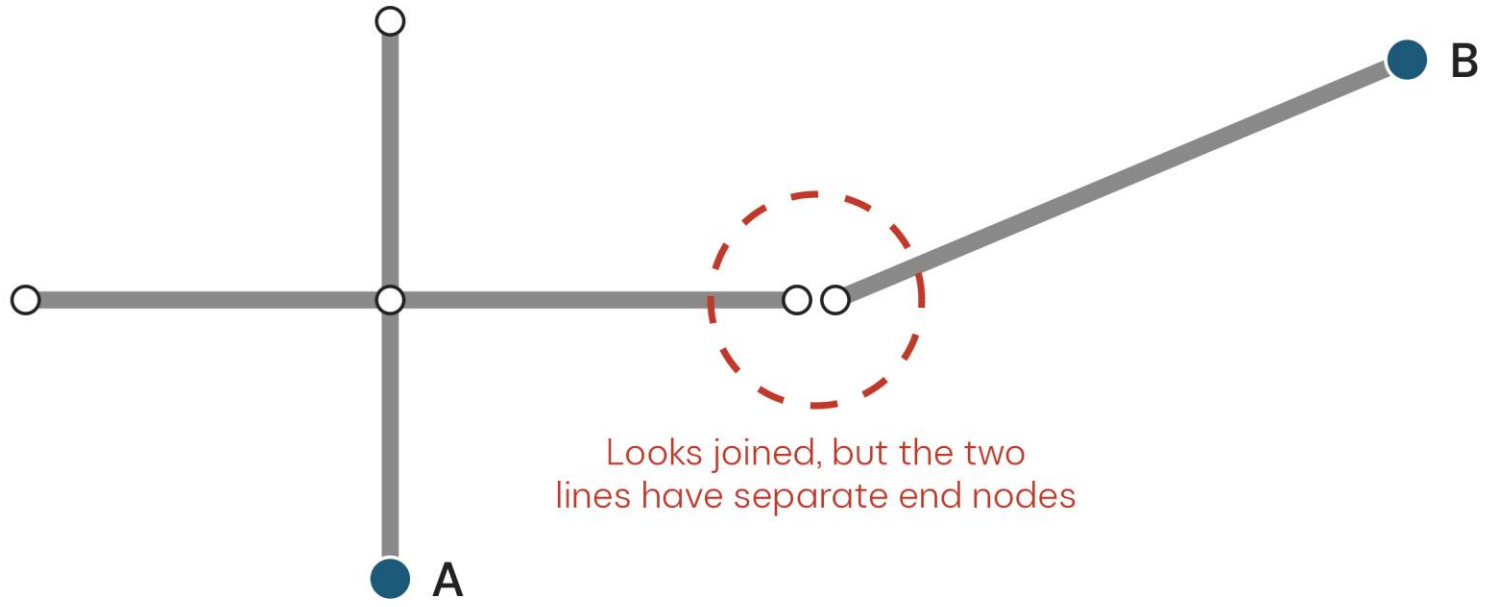
… with Dijkstra’s algorithm we need

- A set of origins (e.g. homes or schools).
- A set of destinations (e.g. shops, jobs or fast-food outlets).
- Some form of a digital spatial network to connect origins and destinations.
- Impedance values per mode of transport / costs for each network segment
- If available: access indicators to construct a weighted graph

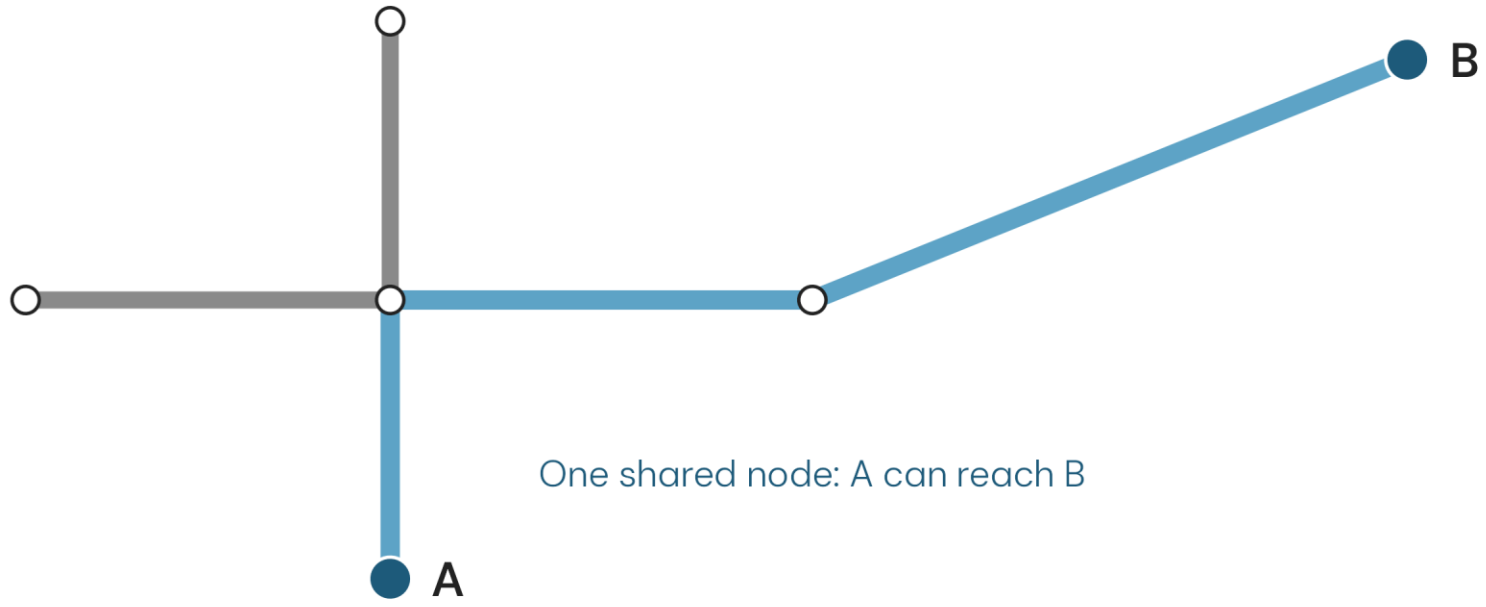
Checking a network: five questions

- Completeness – are all areas covered?
- Attributes – are they correct?
- Connectivity – are all network segments that should be connected, connected?
- Topology – are all network segments connected the way they should be?
- Coverage – is the full network covered?

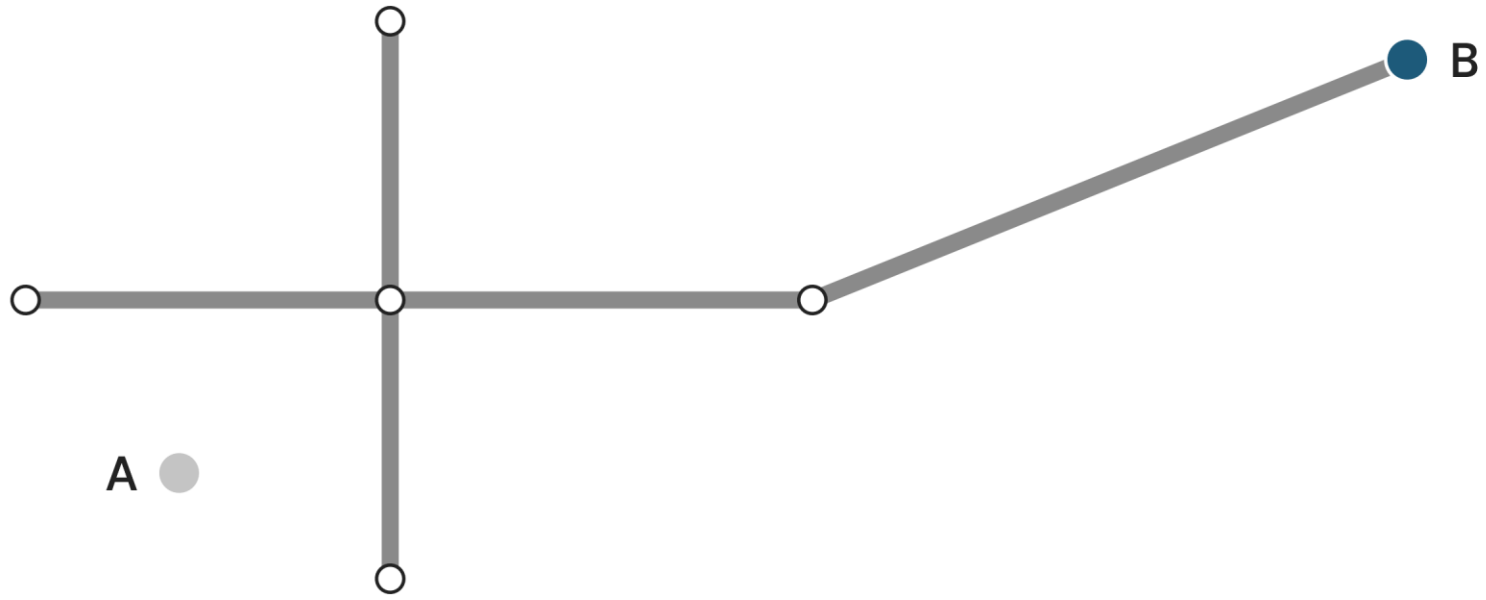
Connectivity: lines that look joined may not be



Fix it with one shared node

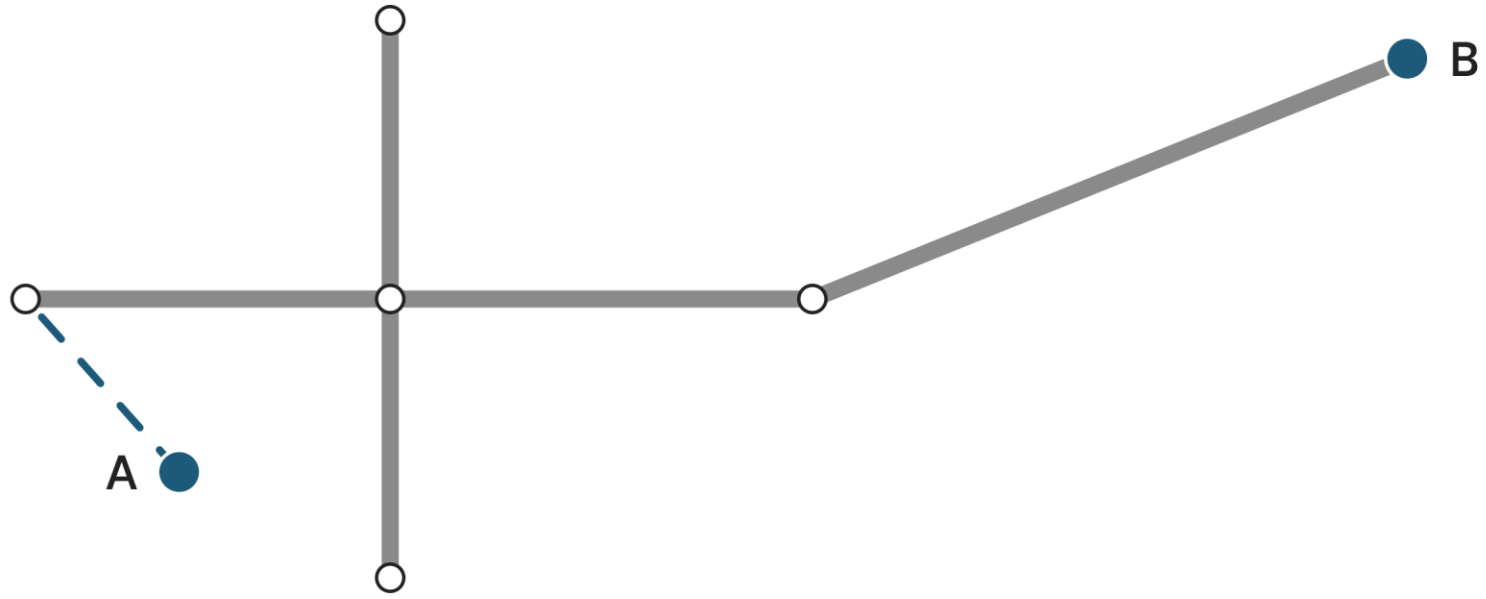


Origins and destinations must be on the network



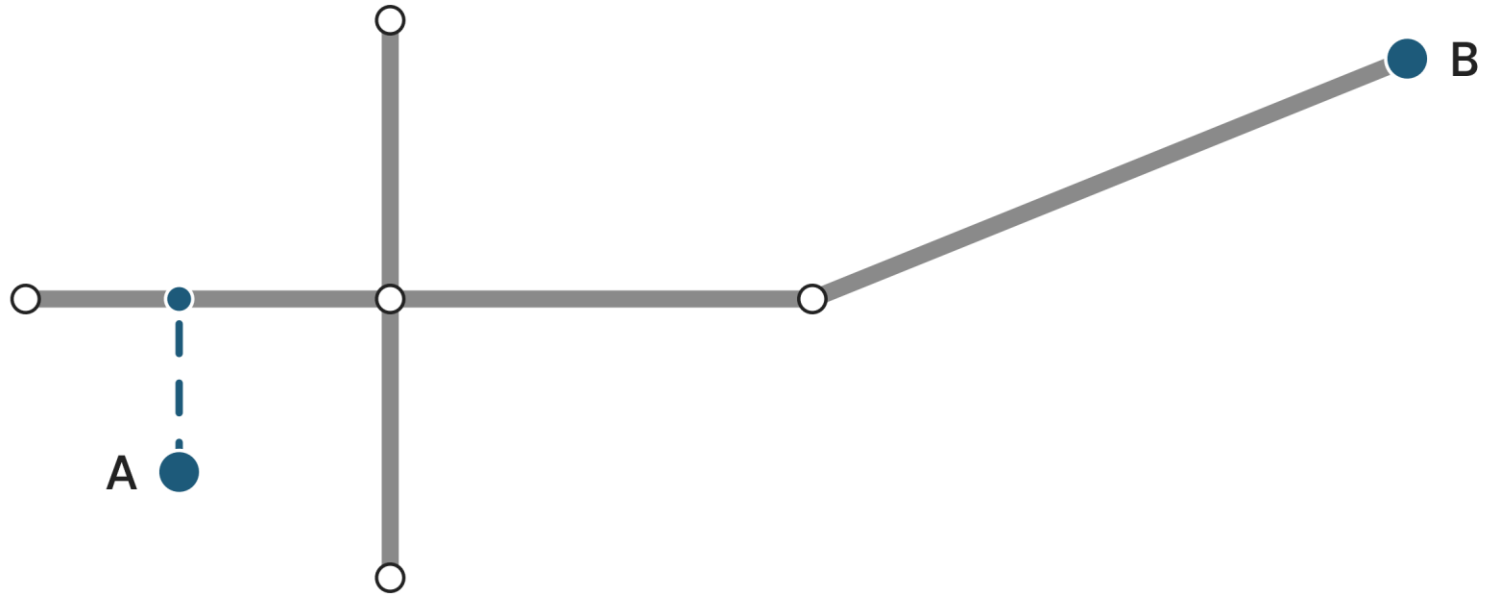
A is not on the network:
there is nothing to route from

Option 1: snap to the nearest node



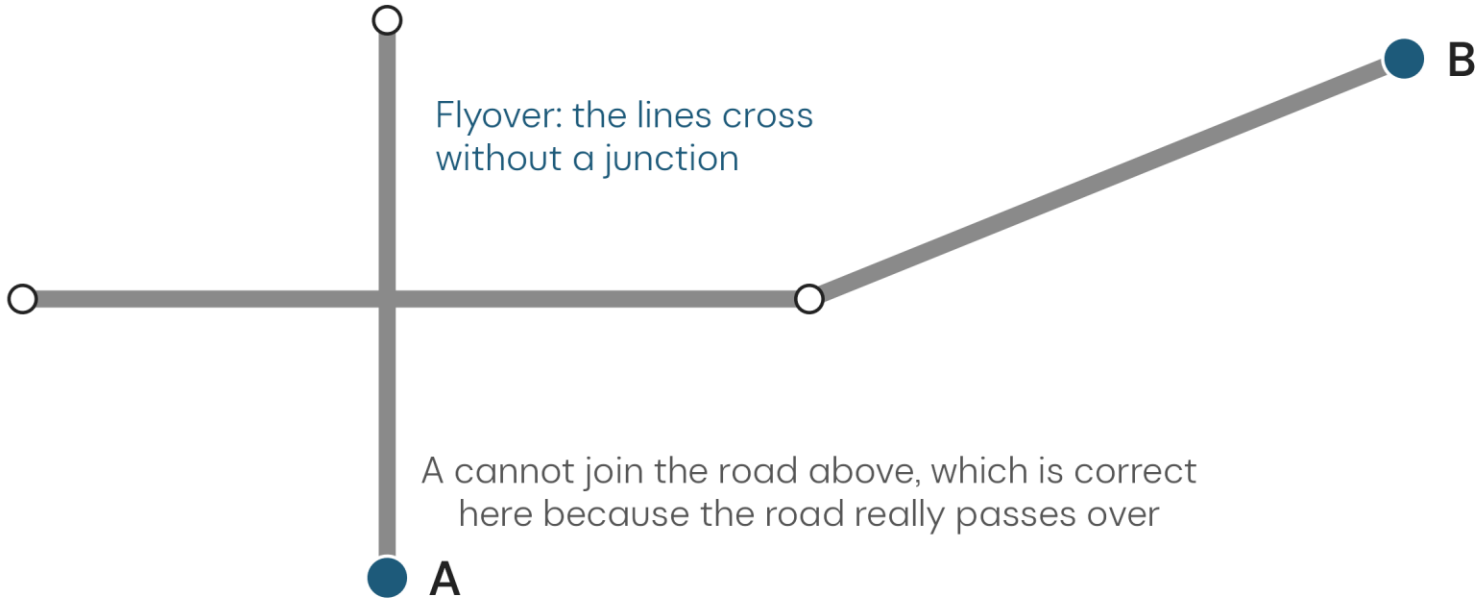
Snap A to the nearest node (120 m away)

Option 2: snap to the street with a feed link

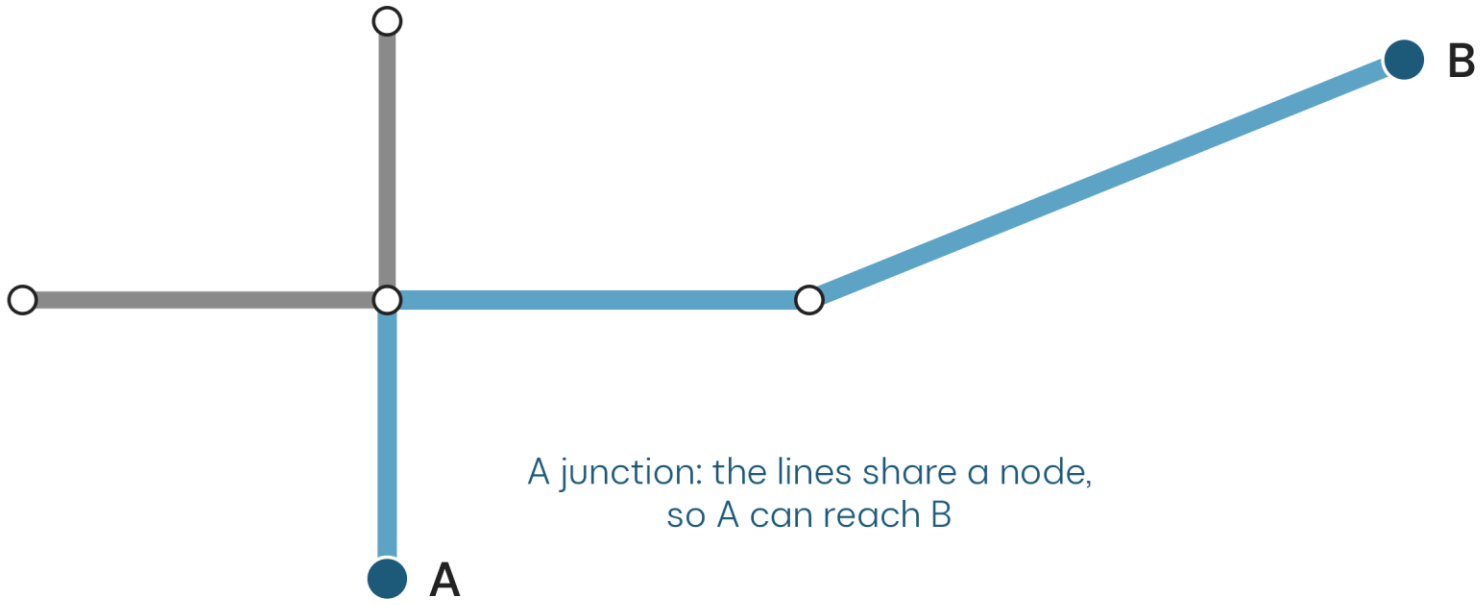


Feed link: a new node on the street, 90 m from A, with its own cost

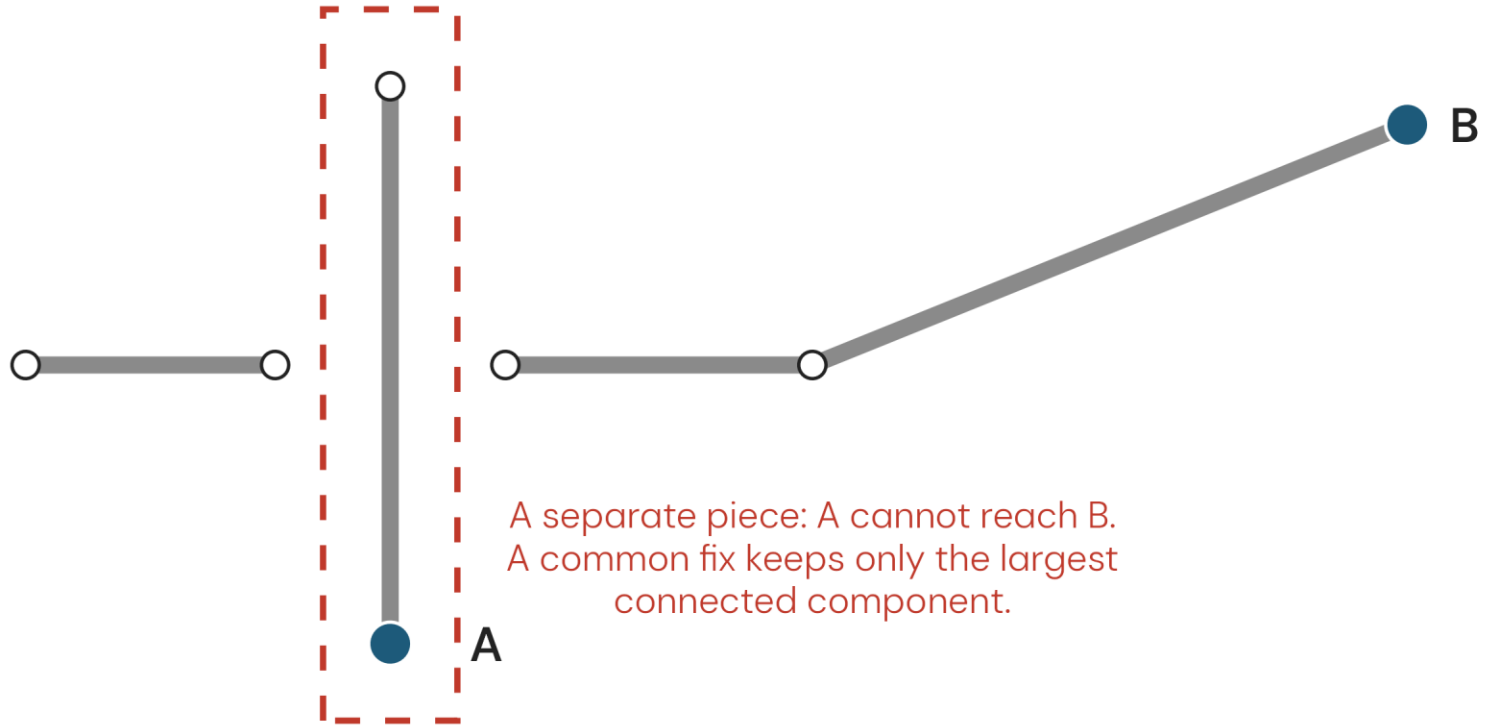
Topology: lines can cross without connecting



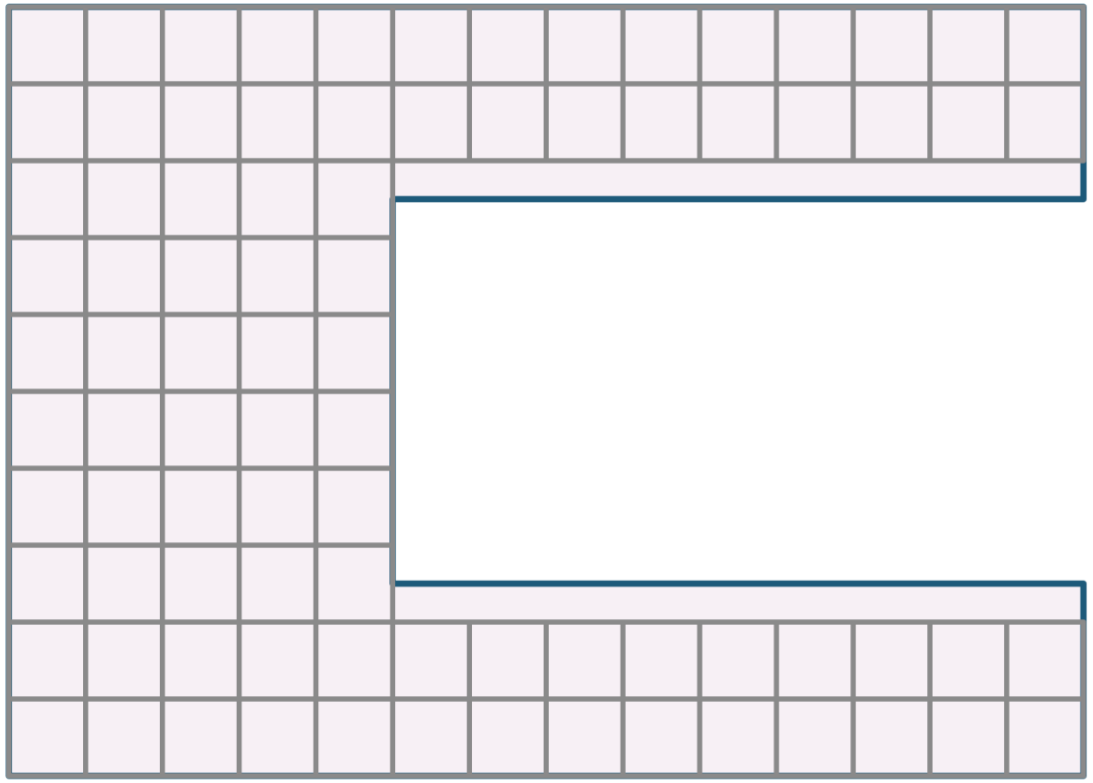
At a junction, the lines must share a node



Disconnected pieces cannot reach each other

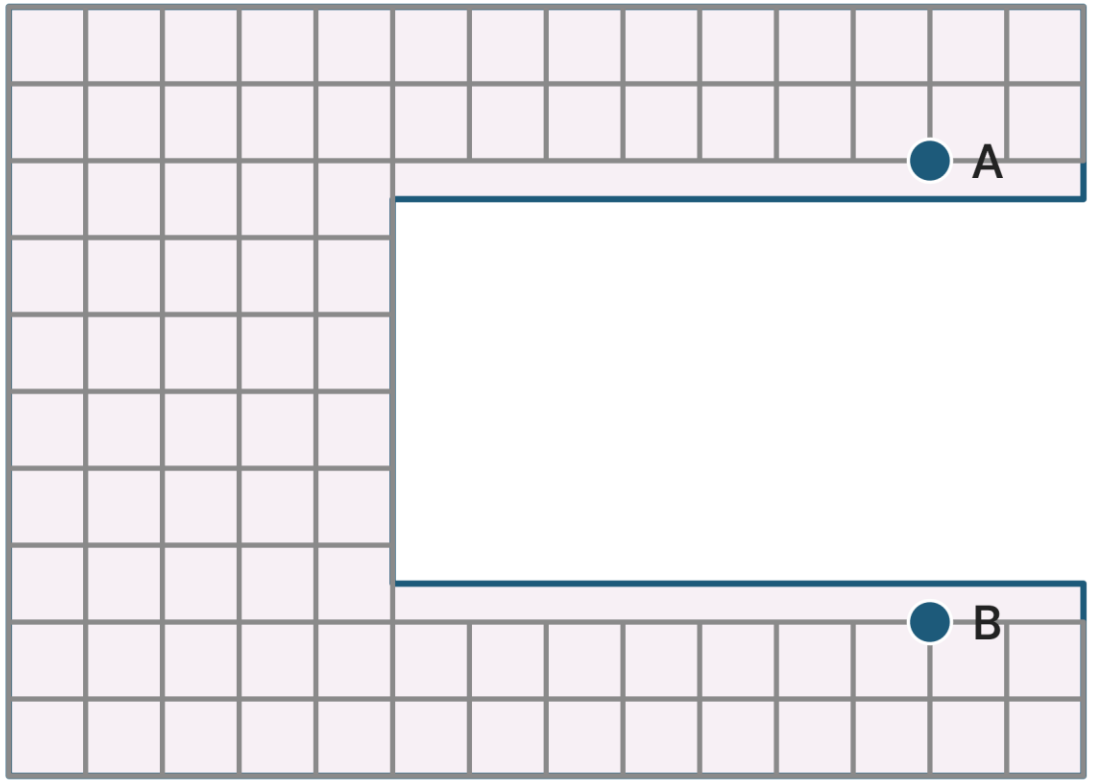


Coverage: does the network cover the study area?



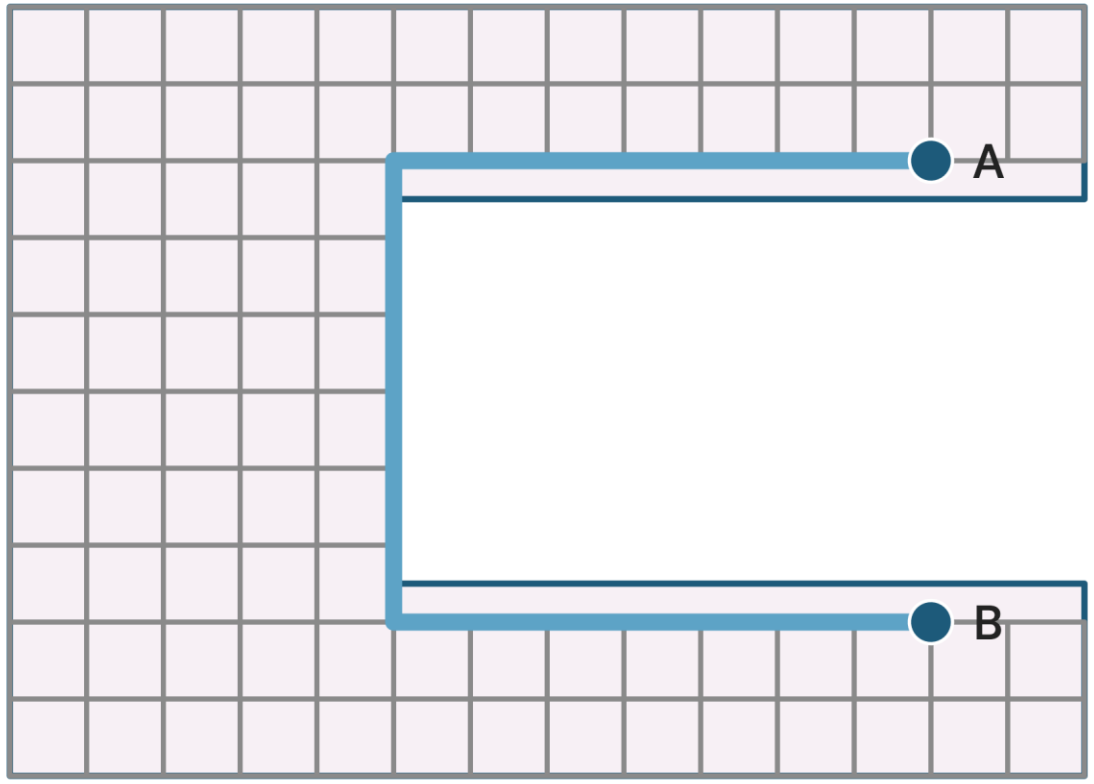
The roads stop at the boundary of the study area

Close locations can be far apart inside the area



A and B are close,
but the study area
wraps around a gap

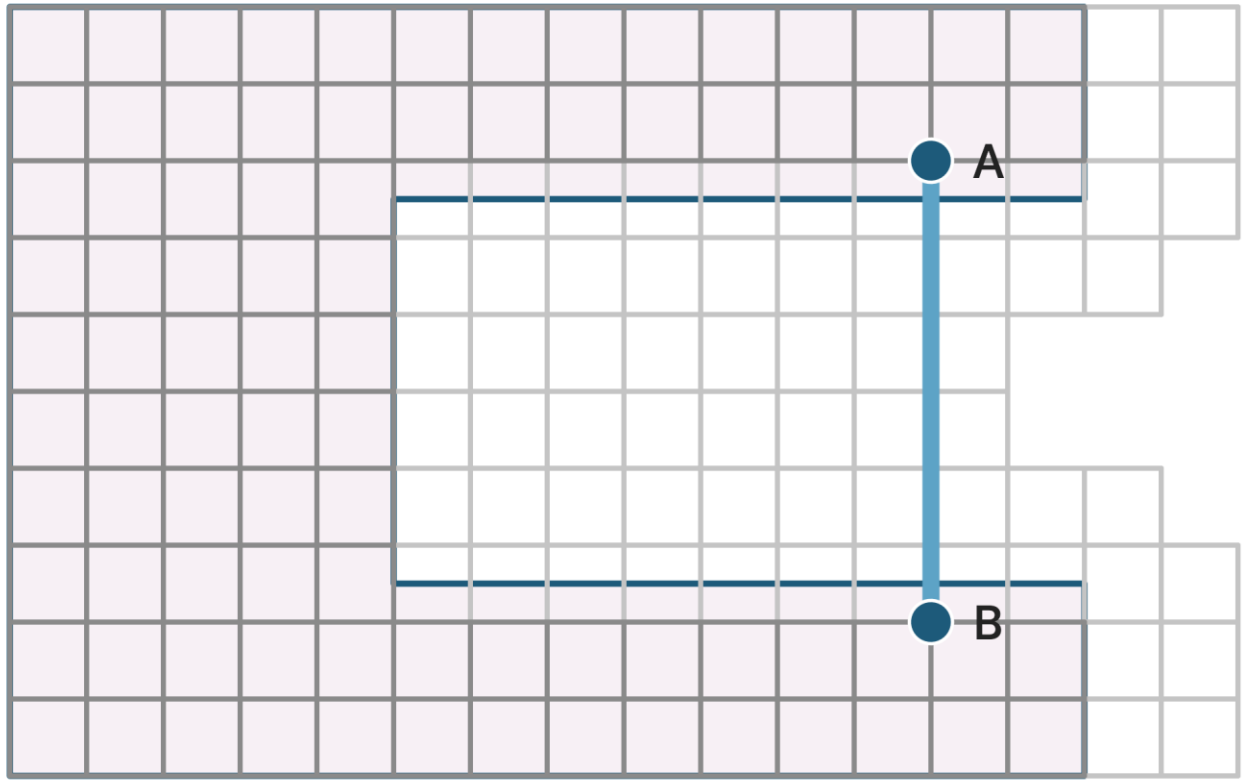
A clipped network forces a long detour



Route from A to B
along the roads inside
the study area

2,000 m

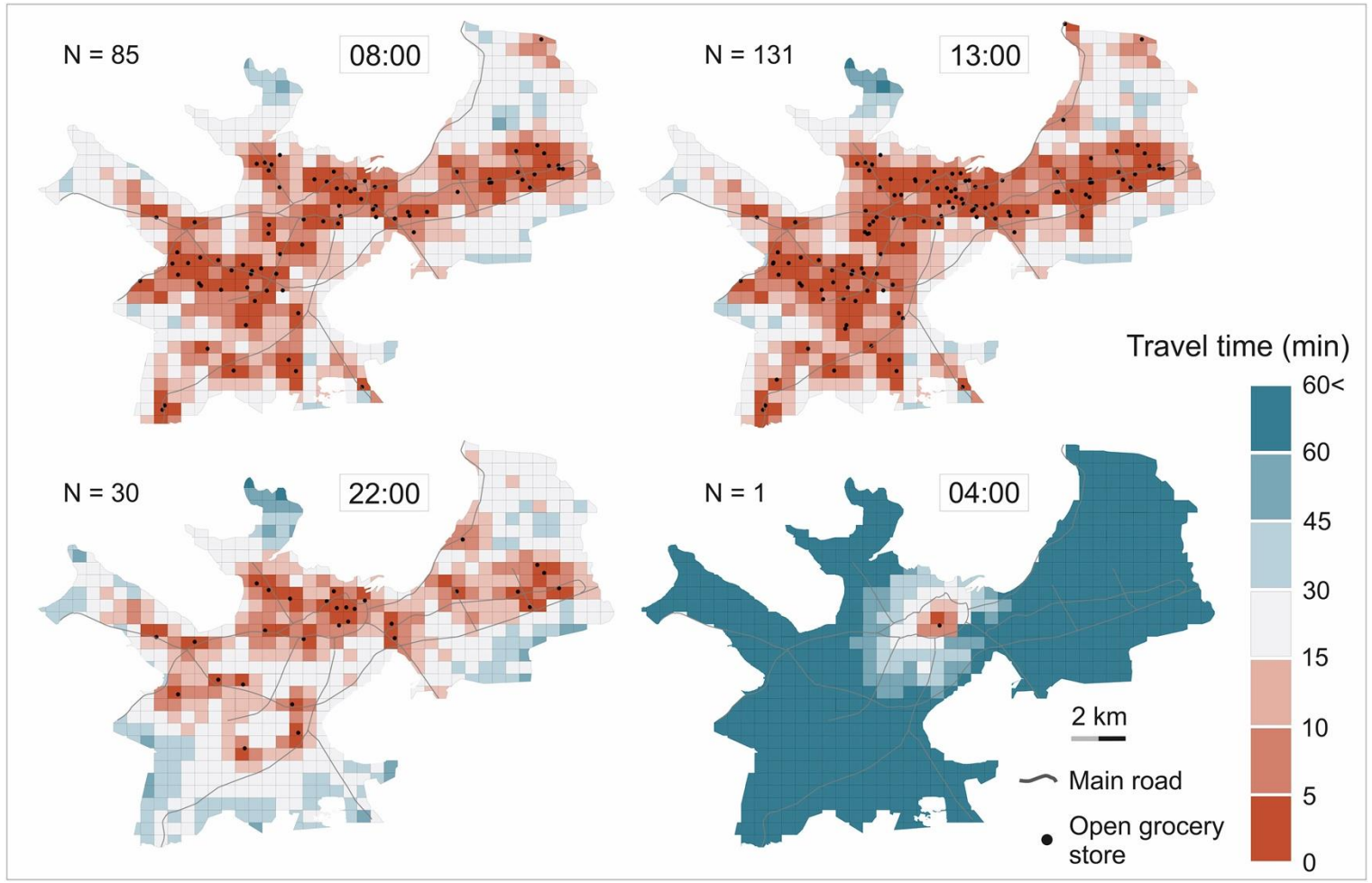
Include roads outside the area



Route when roads up to 250 m outside the area are included

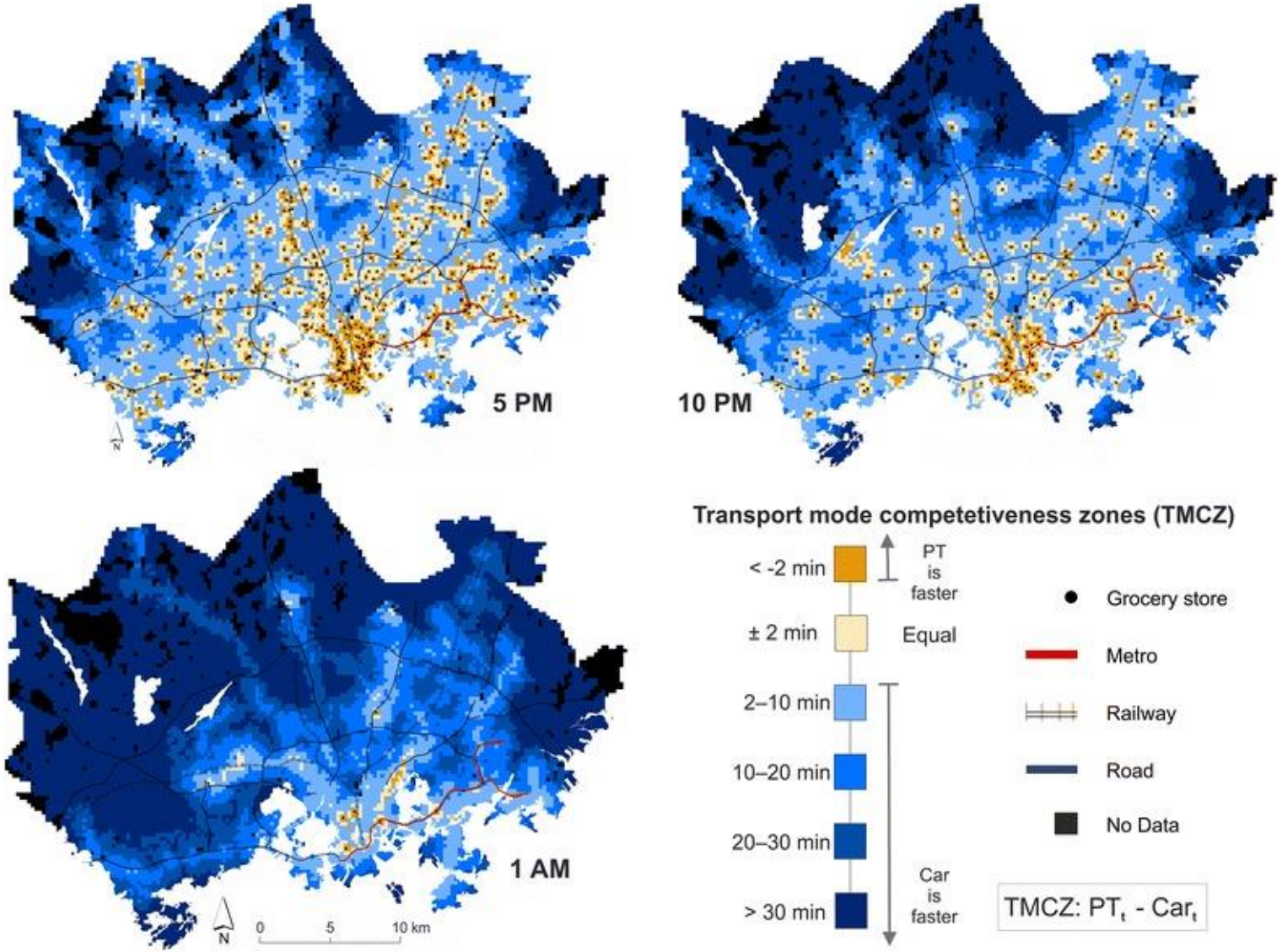
600 m

Accessibility changes through the day



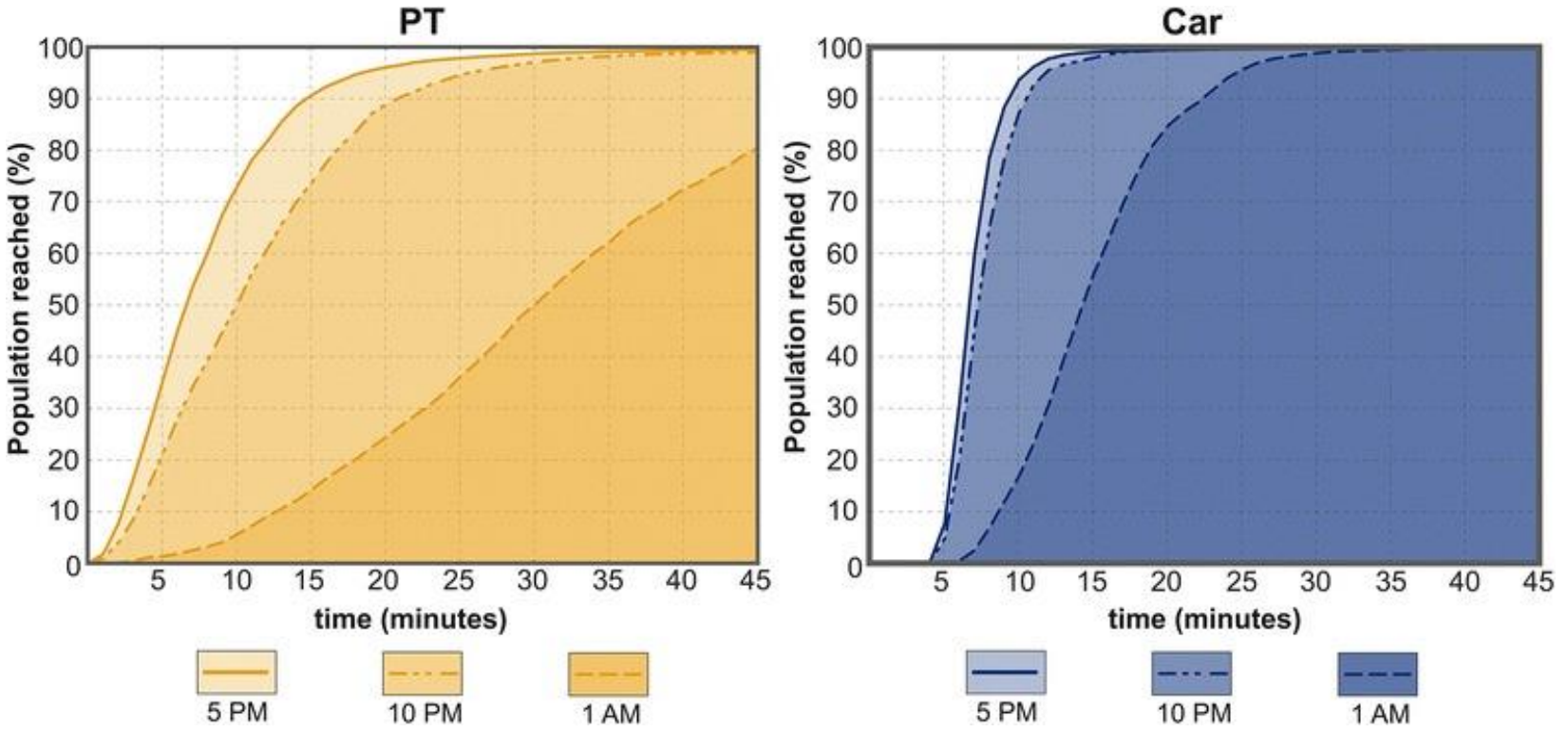
Järv *et al.* 2018

Accessibility depends on the mode of transport



Tenkanen *et al.* 2016

Time of day and mode both matter



Tenkanen *et al.* 2016

Conclusion

- Accessibility needs to be operationalised: accessible to what, for whom and when?
- We need origins, destinations and a network.
- Dijkstra’s algorithm finds the cheapest route from an origin to every node.
- Network quality matters: connectivity, topology and coverage can change the result.
- Open-source tools: dodgr, R5 (r5r, r5py), stplanr.

Questions

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